Track B: The Standard Landscape and the Frontier

7. The standard landscape

The bar machine of Sections 1–6 accepts any chiral algebra as input. Every standard family exhibits two faces simultaneously: the *E-infinity geometric face* (factorization coalgebra on the curve, shadow class in the tautological ring) and the E_1 *algebraic face* (R-matrix profile, dg-shifted Yangian lineage). Physics reads the geometric face; algebra computes the E_1 face; the two are Koszul duals, and the whole monograph runs on the translation between them.

7.0. Three-tier R-matrix classification

The genus-0 binary shadow $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ sorts chiral algebras into three tiers:

- *Tier (a): pole-free commutative.* The Heisenberg algebra $\mathcal{H}_k$ and lattice extensions. The spectral R-matrix is scalar, $R(z) = \exp(k/z)$, and the Yang–Baxter equation is trivial. Shadow class **G**.
- *Tier (b): VA with poles.* Affine Kac–Moody $\widehat{\mathfrak{g}}_k$, Virasoro, principal $\mathcal{W}_N$, $\beta\gamma$, non-principal $\mathcal{W}$-algebras. The classical r-matrix is $k\Omega/z$ (affine) or $(c/2)/z^3 + 2T/z$ (Virasoro) or the DS-reduced analogue. The level prefix is load-bearing: at $k = 0$, the affine r-matrix vanishes. Shadow class **L**, **C**, or **M** depending on the depth of the OPE.
- *Tier (c): genuinely E_1.* The dg-shifted Yangian $Y^{\mathrm{dg}}_{\hbar}(\mathfrak{g})$, Etingof–Kazhdan quantum vertex algebras, line-operator atoms of 3d HT theories. The R-matrix carries genuine topological direction data: nonsymmetric braiding, meromorphic poles of higher order, RTT relations that do not factor through a commutative vertex algebra. These are the nonlocal atoms of the E_1 chiral world; every standard VOA is E_∞ (local), and E_1 primacy requires passing to the tier (c) atoms.

Tier (a) is the free-field archetype, tier (b) supplies the standard landscape, tier (c) is the genuine E_1 frontier occupied in Volumes II and III.

The seven families below exhaust the tier (a)/(b) landscape. Each receives both faces: the geometric face (OPE, bar complex, shadow depth, genus-2 shells) and the algebraic face (R-matrix profile, Koszul dual, Yangian lineage). No two families share all invariants; taken together they realise every depth class $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}$ and every genus-2 shell profile.

7.1. Heisenberg

The Gaussian archetype: shadow depth 2, tower terminates at degree 2, every higher-genus invariant is determined by κ alone.

Geometric face. One bosonic field α of weight 1, $\alpha(z)\alpha(w) \sim k/(z-w)^2$, $c = 1$. The bar complex is $\bar{B}(\mathcal{H}_k) = T^c(s^{-1}\overline{\mathcal{H}_k})$ on the augmentation ideal, with $d_{\bar{B}}$ acting by pairwise contraction into scalars (no simple pole, no Lie bracket). Bar cohomology concentrates: $H^1 = \mathbb{C} \cdot \alpha^*$, $H^n = 0$ for $n \geq 2$. Genus-2 shell profile: only loop∘loop. Shadow class **G**.

Algebraic face. The Koszul dual is the curved symmetric chiral algebra $\mathcal{H}_k^! = (\mathrm{Sym}^{\mathrm{ch}}(V^*), m_0 = -k\omega)$, not a flat VOA. The r-matrix is scalar, $r(z) = k/z^0$ (degree-zero central element), and the spectral R-matrix is $R(z) = \exp(k/z)$ (Tier (a)). Line-operator category $\mathcal{C} = \mathsf{Vect}$: the braiding is symmetric, and $\mathcal{H}_k^!$-modules are ordinary vector spaces.

Modular characteristic. $\kappa(\mathcal{H}_k) = k$, $\kappa(\mathcal{H}_k^!) = -k$, $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = 0$ (the free-field/KM branch of complementarity: sum equals 0 for free-field and affine Kac–Moody families, in contrast to $\kappa + \kappa' = 13$ for Virasoro). The universal MC element is $\Theta_{\mathcal{H}_k} = k \cdot \eta \otimes \Lambda$ with $\eta = d\log(z_1 - z_2)$ and $\Lambda = \lambda_1$; the five-component differential collapses to loop∘loop contractions of the single propagator.

Higher-genus. For uniform weight, the $\hat{A}$-genus controls every F_g:

$$\sum_{g\geq 1} F_g\, x^{2g} \;=\; k\big(\hat{A}(ix) - 1\big) \;=\; k\Big(\tfrac{x/2}{\sin(x/2)} - 1\Big),$$

so $F_1 = k/24$, $F_2 = 7k/5760$, $F_3 = 31k/967680$. The series converges exponentially, $|F_g| \sim 2|k|/(2\pi)^{2g}$.

7.2. Affine Kac–Moody

The Lie/tree archetype: shadow depth 3, cubic shadow is the Lie bracket, Jacobi kills everything higher.

Geometric face. For simple $\mathfrak{g}$ of rank ℓ and dual Coxeter $h^\vee$, the affine vertex algebra $\widehat{\mathfrak{g}}_k$ has currents J^a of weight 1 with

$$J^a(z)J^b(w) \;\sim\; \frac{f^{ab}{}_c\, J^c(w)}{z-w} \;+\; \frac{k\,\kappa^{ab}}{(z-w)^2}.$$

Both pole orders are present: simple pole carries the bracket, double pole carries the level. The bar differential picks up a Chevalley–Eilenberg component from the simple pole and a curvature component from the double pole; $d_{\bar{B}}^2 = 0$ requires the full Borcherds identity, not Jacobi alone. PBW concentration at all genera is proved. Genus-2 shell profile: loop∘loop and sep∘loop active, planted-forest absent. Shadow class **L**.

Algebraic face. The classical r-matrix is $r_{\mathrm{cl}}(z) = k\,\Omega/z$ with $\Omega = \sum_a J^a \otimes J^a$; it vanishes identically at $k = 0$, as it must. The CYBE is the Arnold relation on $\overline{C}_3(\mathbb{C})$ evaluated on the Casimir. The Koszul dual is the affine algebra at the Feigin–Frenkel dual level: $\widehat{\mathfrak{g}}_k^! = \widehat{\mathfrak{g}}_{-k-2h^\vee}$, implementing the involution $k + h^\vee \mapsto -(k + h^\vee)$.

Modular characteristic (KM branch of complementarity: sum equals zero for Kac–Moody and free-field families, in contrast to the Virasoro branch where the sum equals 13).

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee} \qquad (k \neq -h^\vee).$$

KM-branch complementarity: $\qquad \kappa(\widehat{\mathfrak{g}}_k) \;+\; \kappa(\widehat{\mathfrak{g}}_k^!) \;=\; 0 \quad (\text{NOT } 13,\ \text{which is the Virasoro branch}).$

Sugawara and critical level. At $k \neq -h^\vee$, $T = (k+h^\vee)^{-1}\sum :J^aJ^a:/2$ and $c = k\dim\mathfrak{g}/(k+h^\vee)$. At $k = -h^\vee$ the Sugawara construction is *undefined*, $\kappa = 0$, and the Feigin–Frenkel centre $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong$

$\mathrm{Fun}(\mathrm{Op}_{^L\mathfrak{g}}(D))$ is a commutative VA of $^L\mathfrak{g}$-opers on the formal disk. Critical-level chiral Koszul duality is bar-level geometric Langlands.

7.3. Virasoro

The mixed archetype: shadow depth infinity, tower never terminates, the self-coupling $T_{(1)}T = 2T$ drives all nonlinear complexity.

Geometric face. One generator T of weight 2, $T(z)T(w) \sim (c/2)/(z-w)^4 + 2T/(z-w)^2 + \partial T/(z-w)$. Three pole orders. Mode algebra: $[L_m, L_n] = (m-n)L_{m+n} + (c/12)(m^3-m)\delta_{m+n,0}$. The bar differential on $sT|sT$ mixes the order-4 (curvature) and order-2 (self-coupling) poles; at higher tensor degree the bar differential is quadratic in the structure constants. PBW concentration holds. Genus-2 shell profile: all three shells active including planted-forest (first family to see log-FM boundary strata of $\overline{\mathcal{M}}_{2,n}$). Shadow class **M**.

Algebraic face. The r-matrix is $r_c(z) = (c/2)/z^3 + 2T/z$, second-order and zeroth-order poles absorbed from the fourth- and second-order OPE poles by the $d\log$ extraction. The Koszul dual is $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$, self-dual at $c = 13$. The quartic contact shadow is

$$Q^{\mathrm{contact}}_{\mathrm{Vir}} = \frac{10}{c(5c+22)},$$

with poles at $c = 0$ and $c = -22/5$ reflecting the $(2,5)$ minimal-model vacuum degeneracy. The quintic obstruction $o^{(5)} = \{C_{\mathrm{Vir}}, Q^{\mathrm{contact}}\}_H = 480/[c^2(5c+22)]\, x^5$ is nonzero, so no finite truncation captures Virasoro.

Modular characteristic. $\kappa(\mathrm{Vir}_c) = c/2$, $\kappa(\mathrm{Vir}_{26-c}) = 13 - c/2$, and the complementarity constant is

$$\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_c^!) = 13,$$

the unique family where complementarity is nonzero. The constant 13 is half the critical dimension 26, and its nonvanishing reflects the impossibility of cancelling the curvature of a one-generator algebra against the curvature of its dual within the same one-parameter family.

Higher-genus corrections. Beyond the linear curvature:

$$\delta H^{(1)}_{\mathrm{Vir}} = \frac{120}{c^2(5c+22)} \cdot x^2, \qquad \rho^{(1)}_{\mathrm{Vir}} = \frac{240}{c^3(5c+22)}.$$

These are determined by Q^{contact}, not by κ, and are the first nonlinear genus-1 invariants.

Critical dimension $c = 26$. The shadow connection

$$\nabla^{\mathrm{hol}}_c = d - \tfrac{c}{2}\,\omega_g - \tfrac{10}{c(5c+22)}\,\delta_4 - \tfrac{120}{c^2(5c+22)}\,\delta_H - \cdots$$

is flat for every c by the MC equation. The critical dimension arises from the *dual* curvature vanishing: $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2 = 0$ forces $c = 26$. At $c = 26$ the Koszul dual Vir_0 is uncurved; the algebra itself retains $\kappa = 13$. This is the string-theoretic content of the critical dimension.

Polyakov identification. The Polyakov formula $\log(\det'\Delta_{g_1}/\det'\Delta_{g_0}) = -(1/6\pi)\int(|\nabla\sigma|^2 + R\sigma)$ is the degree-2 shadow evaluation at $\kappa = 1$; for general κ on the uniform-weight scalar lane the shadow

$\Theta^{\min}_{\mathcal{A}} = \kappa \cdot \eta \otimes \Lambda$ integrates against σ to reproduce the conformal anomaly with coefficient $\kappa/(6\pi)$. The higher-degree corrections δ_4, δ_H are inaccessible to the free-field path integral.

7.4. $\mathcal{W}_3$

First multi-generator family. Two generators T (weight 2) and W (weight 3) produce cross-channel mixing in the genus-1 Hessian, a 2×2 spectral discriminant, and the first genuine multi-channel genus-2 correction .

Geometric face. The WW OPE:

$$W(z)W(w) \sim \frac{c/3}{(z-w)^6} + \frac{2T}{(z-w)^4} + \frac{\partial T}{(z-w)^3} + \frac{1}{(z-w)^2}\left[\tfrac{16}{22+5c}\Lambda + \tfrac{3}{10}\partial^2 T\right] + \cdots,$$

where $\Lambda =: TT: -(3/10)\partial^2 T$ is the weight-4 quasi-primary. At $c = -22/5$, Λ becomes null and $\mathcal{W}_3$ is not finitely generated by T, W alone. Shadow class $\mathbf{M}$. Genus-2: all three shells active, planted-forest sensitive to $WW\Lambda$ channel.

Algebraic face. The Koszul dual is $\mathcal{W}_3^! = \mathcal{W}_{3,100-c}$, self-dual at $c = 50$. The classical r-matrix is the Casimir piece plus the T- and W-mediated corrections; the per-channel eigenvalues are $\kappa_T = c/2$ and $\kappa_W = c/3$, and the spectral discriminant is $\Delta_{\mathcal{W}_3}(x) = (1 - \frac{c}{2}x)(1 - \frac{c}{3}x)$ with branch points $2/c$ and $3/c$. Line-operator category $\mathcal{C}_c^{W_3}$ is a module category for the dg-shifted W_3-Yangian.

Modular characteristic.

$$\kappa(\mathcal{W}_3) = \frac{5c}{6}, \qquad \kappa(\mathcal{W}_{3c}) + \kappa(\mathcal{W}_{3100-c}) = \frac{250}{3}.$$

The weight-2 generator contributes $c/2$, the weight-3 generator contributes $c/3$, giving $5c/6$ additively.

Genus-2 cross-channel. (all-weight, with cross-channel correction.)

$$F_2(\mathcal{W}_3) = \underbrace{\tfrac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}} + \underbrace{\tfrac{c+204}{16c}}_{\delta F_2^{\mathrm{cross}}},$$

with $\delta F_2^{\mathrm{cross}}$ computed graph-by-graph over the seven genus-2 stable graphs. The scalar formula of the uniform-weight genus expansion *fails* at $g \geq 2$ for multi-weight algebras; $\delta F_g^{\mathrm{cross}}$ accelerates with genus and dominates the scalar part by $g = 4$. No classical matrix-model spectral curve reproduces this tower. A universal closed form for principal $\mathcal{W}_N$:

$$\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = \tfrac{(N-2)(N+3)}{96} + \tfrac{(N-2)(3N^3+14N^2+22N+33)}{24c},$$

exact for $\mathcal{W}_3$ and a lower bound for $N \geq 4$.

Holographic datum.

$$\mathcal{H}(\mathcal{W}_{3,c}) = \big(\mathcal{W}_{3,c},\ \mathcal{W}_{3,100-c},\ \mathcal{C}_c^{W_3},\ r_c^{W_3}(z),\ \Theta_c,\ \nabla_c^{\mathrm{hol}}\big).$$

7.5. $\beta\gamma$ system

The contact archetype. First nonlinearity at degree 4 (not 3); rank-one abelian rigidity kills it; shadow depth 4. The unique standard family where termination has a structural mechanism distinct from Lie cancellation.

Geometric face. Fields β of weight λ, γ of weight $1-\lambda$, with $\beta(z)\gamma(w) \sim 1/(z-w)$ and both self-OPEs vanishing. Simple pole only. Central charge $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$; at $\lambda = 1$, $c = 2$. (Partner: the fermionic bc system has $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$, and $c_{\beta\gamma} + c_{bc} = 0$; verify at $\lambda = 1$: $c_{\beta\gamma} = 2$, $c_{bc} = -2$.)

The cubic shadow $\ell_3^{(0)}$ vanishes because β and γ carry different weights. The quartic $\ell_4^{(0)} \neq 0$, from $\langle\beta\gamma\beta\gamma\rangle$, but the quartic contact invariant $\mu_{\beta\gamma} = \langle\eta, m_3(\eta,\eta,\eta)\rangle = 0$ by the U(1) ghost-number rigidity. Shadow class **C**. Genus-2: loop∘loop only (contact termination, distinct from Gaussian collapse).

Algebraic face (free-field/KM branch of complementarity, sum = 0; contrast Virasoro branch where sum is 13). Koszul dual is the bc ghost system: $(\beta\gamma)^! = bc$, exchanging bosonic and fermionic statistics. $\kappa(\beta\gamma) = c_{\beta\gamma}/2 = 6\lambda^2 - 6\lambda + 1$, and $\kappa(\beta\gamma) + \kappa(bc) = 0$ (free-field branch). The r-matrix is proportional to the identity Casimir; the spectral R-matrix factors through the ghost-number grading.

7.6. Lattice vertex algebras

Arithmetic laboratory. The shadow tower inherits the Gaussian archetype from the Heisenberg factor; the braiding carries the full arithmetic of the lattice. Curvature sees only rank; braiding sees the quadratic form.

Geometric face. For an even lattice Λ of rank N, quadratic form q, 2-cocycle ε,

$$V_\Lambda^{N,q} \;=\; \mathcal{H}_k^{\otimes N} \otimes_{\mathbb{C}} \mathbb{C}[\Lambda]_\varepsilon,$$

with generators the currents α^i and the vertex operators $e^{\lambda\cdot\phi}$. Shadow class **G**, depth 2, loop∘loop only at genus 2.

Algebraic face and modular.

$$\kappa(V_\Lambda^{N,q}) \;=\; \mathrm{rank}(\Lambda) \;=\; N,$$

independent of ε and q. Curvature-braiding orthogonality: the curvature class $[\kappa\cdot\omega_g] \in H^{1,1}(\overline{\mathcal{M}}_g)$ and the braiding representation $\pi_1(\mathrm{Conf}_n(X)) \to \mathrm{Aut}(V_\Lambda^{\otimes n})$ act on complementary factors of the state space. Self-dual lattices (E_8: $N = 8$; Leech: $N = 24$) give holomorphic VAs with trivial modular functor and theta-function partition Θ_Λ/η^N.

Non-simply-laced. B_2, C_2, G_2, F_4 are all class **L**; the short-root correction to the Killing form shifts κ by the lacing number but preserves depth. Langlands duals (B_N, C_N) have distinct κ but share class.

7.7. Yangians and quantum groups

The line-operator algebra. Yangian $Y(\mathfrak{g})$ governs the topological direction of a 3d HT theory; its R-matrix is the genus-0 binary shadow of Θ. Yang–Baxter is the degree-3 MC equation; RTT encodes all degrees.

***R*-matrix as genus-0 binary shadow.**

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}), \qquad R(z) \;=\; 1 + r/z + r^{(2)}/z^2 + \cdots .$$

The classical Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ is the Arnold relation on $\overline{C}_3(\mathbb{C})$ evaluated on the Casimir, and the three terms correspond to the three boundary divisors of $\overline{C}_3(\mathbb{C})$.

MC3 and all-types CG. On the type-A Yangian surface the post-CG problem reduces to a single compact/completed extension packet by (a) chromatic filtration of $K_0(\mathrm{Rep})$ by q-character support, (b) prefundamental Clebsch–Gordan closure $V_n \otimes L^- = \bigoplus_{j=0}^{n} L^-(n-2j)$ at the character level, and (c) Francis–Gaitsgory pro-nilpotent completion. The CG decomposition is extended to all simple types via the Chari–Moura multiplicity-free ℓ-weight property (Theorem ??, Corollary ??); the remaining post-CG completion packets are open beyond type A.

dg-shifted Yangian. The modular dg-shifted Yangian $Y_T^{\mathrm{mod}} = \varprojlim_r Y_T^{\mathrm{mod},\le r}$ is the pro-nilpotent completion of the convolution dg Lie algebra of the 3d HT chiral algebra for T, and the modular R-matrix $R_T^{\mathrm{mod}}(z;\hbar) \in \mathrm{MC}(Y_T^{\mathrm{mod}})$ is an MC element. The Yang–Baxter equation is the degree-3 projection; the higher RTT relations encode all degrees; the inverse limit converges by Mittag-Leffler stabilisation.

Drinfeld–Kohno. The KZ connection $\nabla_{\mathrm{KZ}} = d - \hbar \sum_{i<j} \Omega_{ij}\, dz_{ij}/z_{ij}$ on $\mathrm{Conf}_n(\mathbb{C})$ has monodromy that agrees with the quantum-group representation category by the affine Drinfeld–Kohno theorem; shadow-side this is the statement $\nabla_{\mathrm{KZ}} = \mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k})$ on the affine comparison surface.

7.8. Drinfeld–Sokolov reduction as package morphism

DS: $\widehat{\mathfrak{sl}}_{3k} \longrightarrow \mathcal{W}_{3,c(k)}$ with $c(k) = 2 - 24(k+2)^2/(k+3)$ extends to a morphism of holographic data:

	$\mathcal{H}(\widehat{\mathfrak{sl}}_{3k})$	$\mathrm{DS}(\mathcal{H})$
$\mathcal{A}$	$\widehat{\mathfrak{sl}}_{3k}$	$\mathcal{W}_{3,c(k)}$
$\mathcal{A}^!$	$\widehat{\mathfrak{sl}}_{3-k-6}$	$\mathcal{W}_{3,100-c(k)}$
$r(z)$	$k\,\Omega_{\mathfrak{sl}_3}/z$	$r_c^{W_3}(z)$
κ	$4(k+3)/3$	$5c(k)/6$
depth	3	∞
$\Theta^{(2)}_{\mathrm{pf}}$	0	$\neq 0$

The Casimir r-matrix $k\Omega/z$ specialises to $r_c^{W_3}(z)$; the curvature shifts; the shadow depth jumps $3 \longrightarrow \infty$; the planted-forest shell activates. At the critical level $k = -3$: $\kappa \longrightarrow 0$ on the affine side, $c \longrightarrow \infty$ on the $\mathcal{W}_3$ side (Sugawara pole).

The DS-HPL transfer theorem (Volume II) makes the chain-level mechanism precise. Homological perturbation through the BRST SDR transfers the affine dg-shifted Yangian triple (m, r, Δ) to the $\mathcal{W}$-algebra side: the A_∞ products transfer with nontrivial quartic pole (infinite degree); the r-matrix transfers with the expected pole-order shift; the coproduct is annihilated by a ghost-number obstruction, so $\Delta_z^{\mathcal{W}} = 0$ at all degrees. Gravity does not produce particles, only braids them. Central-charge additivity: the central charge difference $c(\widehat{\mathfrak{sl}}_{3k}) - c(\mathcal{W}_3, k) = 6(4k+5)$ is k-dependent (it includes both the free ghost bc central charge and the BRST stress-tensor improvement).

7.9. Geometric localization principle

Drinfeld–Sokolov reduction is Hamiltonian reduction from $J_\infty(\mathfrak{g}^*)$ to $J_\infty(S_f)$, where $S_f = f + \mathfrak{g}^e$ is the Slodowy transverse slice to the nilpotent orbit of f. The Slodowy slice meets every orbit transversally, carries a Kirillov–Kostant Poisson structure, and its quantisation is the finite W-algebra (Gan–Ginzburg, Premet). The chiral upgrade replaces S_f by its arc space, and $\mathrm{gr}_F\, \mathcal{W}^k(\mathfrak{g}, f) \cong \mathbb{C}[J_\infty(S_f)]$.

In this reading: the PBW–Slodowy collapse is smoothness of the Slodowy slice; the shadow depth jump $3 \to \infty$ under DS records the nonlinearity of the Dirac bracket on $J_\infty(S_f)$; Barbasch–Vogan orbit duality $f \leftrightarrow f^D$ is Springer duality of Slodowy slices, and curvature complementarity is an algebraic identity under the Feigin–Frenkel involution; the master commutative square (Theorem ??) organises the whole picture, with bar complexes on the upper row and Hamiltonian reduction on arc spaces on the lower row.

8. Arithmetic of the shadow obstruction tower

The shadow obstruction coefficients for the standard families are rational functions of the central charge. Their denominators, zeros, and growth rates carry number-theoretic information connecting the shadow calculus to L-functions. What arithmetic does the MC equation force?

The Dirichlet–sewing lift

The genus-1 connected free energy $F_1^{\mathrm{conn}}(\mathcal{A}; q) = -\log\det(1 - K_q)$ encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The Dirichlet–sewing lift

$$S_{\mathcal{A}}(u) \;=\; \sum_{N\geq 1} a_{\mathcal{A}}(N) N^{-u} \;=\; \frac{(2\pi)^u}{\Gamma(u)} \int_0^\infty F_{\mathcal{A}}^{\mathrm{conn}}(e^{-2\pi y})\, y^{u-1}\, dy$$

is a Dirichlet series whose analytic structure is controlled by $\Theta_{\mathcal{A}}$. The sewing amplitudes define an inner product $\langle f, g\rangle_{\mathcal{A}} = \sum_N a_{\mathcal{A}}(N) f(N) g(N)$ with reproducing kernel $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s+t)$; the surface moment matrix $\mathcal{M}_{ij} = K_{\mathcal{A}}(\alpha/2 + i, \alpha/2 + j)$ is positive definite for all $\alpha \geq 2$ (Theorem ??), a Gram matrix $V^T D V$ with Vandermonde V.

Explicit values:

$$\begin{aligned} S_{\mathcal{H}}(u) &= \zeta(u)\zeta(u+1), \\ S_{V_k(\mathfrak{g})}(u) &= \dim(\mathfrak{g}) \cdot \zeta(u)\zeta(u+1), \\ S_{\mathrm{Vir}}(u) &= \zeta(u+1)(\zeta(u) - 1), \\ S_{\mathcal{W}_N}(u) &= \zeta(u+1)\Big((N-1)\zeta(u) - \textstyle\sum_{j=1}^{N-1} H_j(u)\Big), \end{aligned}$$

where $H_j(u) = \sum_{n=1}^{j} n^{-u}$ are partial harmonic sums.

The three-tier Euler–Koszul classification

The *Euler defect* $D_{\mathcal{A}}(u) := S_{\mathcal{A}}(u)/(|W(\mathcal{A})|\, \zeta(u)\, \zeta(u+1))$ sorts chiral algebras:

Tier	Condition	Families
Exact Euler–Koszul	$D_{\mathcal{A}}(u) \equiv 1$	$\mathcal{H}, V_k(\mathfrak{g}), \beta\gamma$
Finitely defective	$1 - D_{\mathcal{A}}(u)$ finite harmonic ratio	Virasoro, $\mathcal{W}_N$
Non-Euler–Koszul	$S_{\mathcal{A}}$ decomposes into Hecke L-functions	lattice VOAs with $h(D) \geq 2$

The Euler–Koszul class $\mathrm{ek}(\mathcal{A}) = \max_i(w_i - 1)$ and the shadow depth $r_{\max}(\mathcal{A})$ are independent invariants (Theorem **??**): character arithmetic and OPE interaction are logically independent axes.

The two-variable L-object

Rankin–Selberg unfolding against $E^*(\tau, s)$ gives

$$L_{\mathcal{A}}(s, u) \;=\; \int_{\mathcal{F}}^{\mathrm{reg}} F_{\mathcal{A}}^{\mathrm{conn}}(\tau) E^*(\tau, s) y^{u-2}\, dx\, dy.$$

The *sewing–Hecke reciprocity theorem* collapses both variables to one: $L_{\mathcal{A}}(s, u) = \Gamma(v)/(2\pi)^v\, S_{\mathcal{A}}(v)$ with $v = s + u - 1$. The scattering poles of E^* at $s = \rho/2$ (nontrivial zeros ρ of ζ) are *not* poles of $L_{\mathcal{A}}$; the nontrivial zeros appear as structural obstructions to the Euler product, not as poles of the L-object.

Shadow-moduli resolution

The shadow-moduli map $t\colon \mathfrak{H} \to \mathbb{C}$ solves $\prod_j (1 - \lambda_j\, t(q))^{c_j} = \det(1 - K_q^{\mathrm{conn}})$ with spectral atoms λ_j and multiplicities c_j. The connected free energy factorises as $F_1^{\mathrm{conn}}(q) = G_{\mathcal{A}}(t(q))$ with

$$G_{\mathcal{A}}(t) \;=\; \int \log(1 - \lambda t)\, d\rho_{\mathcal{A}}(\lambda),$$

determined by the spectral measure alone (Theorem **??**). The moments $\mu_r = \int \lambda^r\, d\rho_{\mathcal{A}}$ satisfy the MC recursion

$$\mu_{r+1} \;=\; (r+1)\, \mathrm{Br}_{r+1}(\mu_2, \ldots, \mu_r; \mathcal{A})$$

degree by degree. For two Satake atoms α, β, the Newton relations give the power sums $p_r = e_1 p_{r-1} - e_2 p_{r-2}$, encoding the MC recursion in spectral coordinates.

Interacting Gram matrix and resonance

The shadow obstruction tower provides corrections to Gram positivity. The interacting weight $w_{\mathcal{A}}(N; \alpha) = a_{\mathcal{A}}(N) \cdot \prod_j (1 - \lambda_j N^{-\alpha/2})^{c_j}$ is unconditionally positive when all atoms λ_j are negative. For Virasoro with $c > 0$: $\lambda_{\mathrm{eff}} = -6/c < 0$, so positivity holds unconditionally for $\alpha \geq 2$. The resonance locus $\mathcal{R}$ is empty for $c > 0$ and lies in $c < 0$, with the Lee–Yang point $c = -22/5$ as the first entry.

The Weil analogy

Weil proof for $C/\mathbb{F}_q$	**MC equation for $\mathcal{A}$**
Curve $C/\mathbb{F}_q$	chirally Koszul $\mathcal{A}$
Frobenius Frob_q	MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$
$\mathrm{Frob}^2 = q$	$D_{\mathcal{A}}^2 = 0$
Étale cohomology $H^i_{\text{ét}}$	cyclic deformation $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$
Weight filtration	degree filtration $\Theta_{\mathcal{A}}^{\leq r}$
Eigenvalues α_i on H^1	spectral atoms λ_j
$\|\alpha_i\| = q^{1/2}$	$\lvert a_{f_j}(p)\rvert \leq 2p^{(k-1)/2}$
Lefschetz L	bracket with $\kappa(\mathcal{A})$

For even unimodular lattice VOAs V_Λ with $\Theta_\Lambda = c_E E_{r/2} + \sum_j c_j f_j$, the Hecke–Newton closure gives: MC equation $\Rightarrow$ prime-local CG closure $\Rightarrow$ CPS automorphy $\Rightarrow$ $\mathrm{Sym}^{r-1}(f)$ $\Rightarrow$ Ramanujan bound. Deligne's theorem supplies the bound itself; the MC framework supplies the systematic Hecke decomposition.

Polyakov–bar-cobar dictionary

Polyakov	Bar-cobar
Conformal anomaly $(c/12)R$	modular characteristic κ
$\det'_\zeta \Delta$	Fredholm $\det(1-K)$
Critical dimension $c = 26$	dual curvature $\kappa(\mathrm{Vir}_{26-c}) = 0$
Ghost system bc	Koszul dual $\mathcal{A}^!$ (BRST pairing)
Genus expansion $Z = \sum g_s^{2g-2} Z_g$	MC5 sewing, $F_g = \kappa\lambda_g^{\mathrm{FP}}$ (uniform)
BPZ equations	shadow connection on comparison surface
Instanton sectors	Novikov-completed bar $B^\Lambda(\mathcal{A})$
Wilson line $\langle W(C)\rangle$	Swiss-cheese boundary module

Anomaly, genus expansion, duality, critical dimension, and non-perturbative completions all descend from $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. The four depth classes **G**/**L**/**C**/**M** correspond to increasing departures from the Gaussian path integral.

Arithmetic packet connection

Hecke operators T_p act on the space $M_{\mathcal{A}}$ of activated graph amplitudes $\ell_\Gamma^{(g)}$. Decompose $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$ into Hecke generalised eigenspaces with eigenvalue $a_\chi(p)$ and nilpotent part N_χ; attach to each eigenspace its packet function $\Lambda_\chi(s)$ (completed L-function of the eigenform). The *arithmetic packet connection*

$$\nabla^{\mathrm{arith}}_{\mathcal{A}} = d - \bigoplus_\chi \frac{\Lambda'_\chi(s)}{\Lambda_\chi(s)}(\mathrm{id}_{M_\chi} + N_\chi)\, ds$$

is flat (each block is scalar form times constant endomorphism). The singular divisor $D_{\mathcal{A}}$ is the zero-pole set of the packet functions, depending only on the arithmetic skeleton $M^{\mathrm{ss}}_{\mathcal{A}}$.

The Eisenstein scattering function on $\mathcal{M}_{1,1}$ is $\varphi(s) = \Lambda^*(1-s)/\Lambda^*(s)$ with $\Lambda^*(s) = \pi^{-s}\Gamma(s)\zeta(2s)$. The frontier defect form $\Omega_{\mathcal{A}} = d\log\Lambda_{\mathrm{Eis}}(s) - d\log\varphi(s)$ vanishes iff the Eisenstein block is gauge-equivalent to the scattering connection. For lattice VOAs at rank r, $\Omega_{V_\Lambda} \neq 0$ in general because $\zeta(s)$ and $\zeta(2s)$ have different zero sets.

Finite Miura defect

The principal $\mathcal{W}_N$ character is $\chi_{\mathcal{W}_N}(q) = \prod_{n\geq 2}(1-q^n)^{-\min(n-1,N-1)}$; the Heisenberg character is $\chi_{\mathcal{H}}(q) = \prod_{n\geq 1}(1-q^n)^{-1}$. The ratio is finite:

$$\chi_{\mathcal{W}_N}(q) = \chi_{\mathcal{H}}(q)^{N-1} \cdot \prod_{m=1}^{N-1}(1-q^m)^{N-m}.$$

The finite product is the Miura defect; all genus-1 arithmetic content in $\mathcal{W}_N$ beyond Heisenberg is carried by it. The prime-side Dirichlet analogue is $D^{\mathrm{prime}}_{\mathcal{W}_N}(u) = -\zeta(u+1)\sum_{m=1}^{N-1}(N-m)m^{-u}$; the packet connection splits into $(N-1)$ Heisenberg copies plus a defect governing modes $m = 1, \ldots, N-1$.

Quartic closure programme

The quartic resonance class $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A})$ satisfies a clutching law along every separating stratum

$$\xi^*\mathfrak{R}^{\mathrm{mod}}_{4,g,n} \;=\; p_1^*\mathfrak{R}^{\mathrm{mod}}_{4,g_1,n_1+1} + p_2^*\mathfrak{R}^{\mathrm{mod}}_{4,g_2,n_2+1} + \mathfrak{T}_{3,\xi}$$

(Theorem ??). The 3×3 Schur complement of the mixed (s, u)-transform extracts Q^{ct}_v on the potential side and its reciprocal on the Gram side. For Virasoro: $\Sigma^{\mathrm{Vir}}_2 = 10/[c(5c + 22)]$.

Three closure conjectures form a hierarchy (each implies its predecessors): *quartic closure* (intersection of Gram loci equals $\{\Re\rho = 1/2\}$, Conjecture ??), *Beilinson closure* (global discrepancy $\mathcal{B}^{\mathrm{glob}}(\rho)$ vanishes iff $\Re\rho = 1/2$, Conjecture ??), and *clutching closure* (residue clutching defect $K_{\mathcal{A},\rho,\xi}$ vanishes for all families iff $\Re\rho = 1/2$, Conjecture ??). The chain of constructions $\Theta_{\mathcal{A}} \to w_{c,\rho,u_0} \to$ Gram matrix $\to$ Schur complement $\to$ quartic resonance divisor $\to$ residue clutching defect $\to$ closure conjecture is proved up to the final link, which is conjectural.

Constrained Epstein zeta

For a lattice VOA V_Λ of rank r, the constrained Epstein zeta $\varepsilon^c_s(V_\Lambda) = \sum'_{\ell\in\bar{\Lambda}} \|\ell\|^{-2s}$ restricts the Epstein series to the sublattice $\bar{\Lambda} \subset \Lambda$ determined by the bar complex. For the rank-1 lattice at self-dual radius $R = 1$, every nonzero integer contributes: $\varepsilon^1_s(R{=}1) = 4\zeta(2s)$. The simplest lattice VOA's constrained Epstein zeta is the Riemann zeta.

The Hecke decomposition: $\varepsilon^r_s(V_\Lambda) = \sum_j c_j L(s, \chi_j)$. Shadow depth d contributes $d - 1$ critical lines to the meromorphic continuation, one for each degree stratum beyond the Gaussian base. The nontrivial zeros of the component L-functions are the scattering resonances of the hyperbolic Laplacian on $\mathcal{M}_{1,1}$: frequencies at which the genus-1 sewing amplitude fails to decay.

Genus-2 Böcherer bridge. For the Leech lattice VOA, the Siegel modular form $\chi_{12} = \mathrm{SK}(f_{22})$ is the Saito–Kurokawa lift of the unique weight-22 cuspidal eigenform, and the MC-derived second Fourier coefficient $c_2 \approx -1.918 \times 10^{-6} \neq 0$ is the first nonzero central L-value produced by the shadow obstruction tower, connecting $\Theta_{\mathcal{A}}$ to genus-2 Siegel modular forms via the Furusawa–Morimoto theorem.

10. The open/closed world (Volume II)

The bar complex lives naturally in three dimensions. On $\mathbb{C}_z \times \mathbb{R}_t$, the holomorphic direction produces the closed-string bar differential of Sections 1–9, extracting OPE residues from collisions in the holomorphic plane. The topological direction produces the open-string coproduct, splitting an ordered tensor sequence at a cut point of the real line. The two are unified by the *holomorphic-topological Swiss-cheese operad* $\mathrm{SC}^{\mathrm{ch,top}}$: closed colour ch $= \overline{C}_k(\mathbb{C})$ governs holomorphic factorization, open colour top $= E_1(m)$ governs topological factorization, and the open/closed mixed sector encodes the action of bulk on boundary. Without this operadic habitat, the bar coproduct has no definition and the line-operator category has no foundation.

The primitive object of 3d HT quantum field theory on $\mathbb{C}_z \times \mathbb{R}_t$ is not the bar complex but the open/closed factorization dg-category C on the bordified curve $\widetilde{X}_D$. Objects are boundary conditions, morphisms are open-string states, and a vacuum b produces a boundary algebra $A_b = \mathrm{End}_C(b)$ only up to Morita equivalence. The governing operad is $\mathrm{SC}^{\mathrm{ch,top}}$; it is homotopy-Koszul (proved via Kontsevich formality), so the bar-cobar adjunction on $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is a Quillen equivalence.

The bar complex of Volume I is the ordered E_1 coalgebraic engine for C: its differential records holomorphic collision data, and its coproduct records ordered topological splitting. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure itself lives on the derived-center pair $(C^\bullet_{\mathrm{ch}}(A_b, A_b),\ A_b)$. Three objects must never be conflated:

(i) the *bar complex* $\bar{B}(A_b)$ classifies twisting morphisms, universal couplings between A_b and $A^!_b$;

(ii) the *chiral derived centre* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(C) = R\,\mathrm{Hom}_{\mathrm{Fun}(C,C)}(\mathrm{Id}, \mathrm{Id})$ is the universal bulk, computed by chiral Hochschild cochains in any chart;

(iii) the Koszul dual $A^!_b$ governs the line-operator category $A^!_b$-mod.

The primitive hierarchy

$$C_{\mathrm{op}} \xrightarrow{\mathrm{End}(b)} A_b \xrightarrow{\bar{B}} \bar{B}(A_b) \xrightarrow{\Theta} \Theta_{\mathcal{A}} \xrightarrow{+\mu^{M_j}} \Theta^{\mathrm{oc}}_{\mathcal{A}}$$

records successive lossy functors: C_{op} is Morita invariant; A_b depends on b; $\bar{B}$ forgets the bulk; $\Theta_{\mathcal{A}}$ is the closed-sector MC element; $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ restores open/closed data. Modularity is determined by the open sector: a cyclic trace on $\mathrm{End}(b)$ seeds the Calabi–Yau structure, the annulus identification $\int_{S^1} C \simeq \mathrm{HH}_\bullet(C)$ is the first modular shadow, and clutching along boundary circles raises genus.

Volume II proceeds in four stages. Stage 1 (local one-colour theorem: Swiss-cheese universality of the derived centre), Stage 2 (open primitive: C with Morita invariance), and Stage 3 (globalisation on tangential log curves (X, D, τ)) are proved. In Stage 4 (modularisation), the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is constructed, the modular MC equation holds at all genera, and the projection principle recovers both the closed-sector genus tower and the open-sector module structures. The annulus trace is proved; the independent higher-genus derivation and chain-level cooperad compatibilities remain open.

10.1. The Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$

Two colours $\{\mathrm{ch}, \mathrm{top}\}$, with operation spaces

$$\begin{aligned}
\mathrm{SC}(\mathrm{ch}^k; \mathrm{ch}) &= \overline{C}_k(\mathbb{C}),\\
\mathrm{SC}(\mathrm{ch}^k, \mathrm{top}^m; \mathrm{top}) &= \overline{C}_k(\mathbb{C}) \times E_1(m),\\
\mathrm{SC}(\dots, \mathrm{top}, \dots; \mathrm{ch}) &= \varnothing.
\end{aligned}$$

Closed-to-closed is the Fulton–MacPherson compactification in $\mathbb{C}$; the mixed direction combines holomorphic configurations acting on the fibre above a topological interval; open-to-closed is forbidden. Operadic composition is the boundary stratification of FM spaces in the holomorphic direction and nested interval inclusion in the topological direction.

A bar element of tensor degree k lives on the product $\overline{C}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. The bar differential extracts OPE residues along collision divisors of the first factor: when points z_i, z_j collide, the residue of $\eta_{ij} = d\log(z_i - z_j)$ yields the mode $a_{(n)}b$. The coproduct splits the ordered sequence at a cut point of the second factor: for $t_1 < \cdots < t_k$ with $t_p < c < t_{p+1}$, $(a_1 \otimes \cdots \otimes a_k)$ decomposes as $(a_1 \otimes \cdots \otimes a_p) \otimes (a_{p+1} \otimes \cdots \otimes a_k)$. The bar complex with both structures is a $\mathrm{SC}^{\mathrm{ch,top}}$-algebra.

10.2. Homotopy-Koszulity

The classical Swiss-cheese operad SC is Koszul (Livernet–Voronov, Ginzburg–Kapranov), with quadratic presentation and distributive-lattice criterion on the poset of nested disks. Kontsevich formality supplies a quasi-iso $\Phi\colon \mathrm{SC}^{\mathrm{ch,top}} \xrightarrow{\sim} \mathrm{SC}$ from configuration-space integrals on FM spaces with propagators, extended by including $\mathbb{H}$ and $\mathbb{R}$. The bar-cobar adjunction preserves quasi-isos, and two-out-of-three in the operadic model category gives $\Omega\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}}) \xrightarrow{\sim} \mathrm{SC}^{\mathrm{ch,top}}$. The bar-cobar adjunction on $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is therefore a Quillen equivalence, with three consequences:

(i) every $\mathrm{SC}^{\mathrm{ch,top}}$-algebra admits a cofibrant bar-cobar resolution;
(ii) the homotopy category of $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is equivalent to that of coalgebras over the Koszul dual cooperad;
(iii) the line-operator category $A_b^!$-mod is unconditionally defined without finiteness hypotheses on $\mathcal{A}$.

10.3. The A_∞ chiral algebra structure

The full Swiss-cheese MC element $\alpha_T \in \mathrm{MC}(\mathfrak{g}_T^{\mathrm{SC}})$ lives in the convolution algebra $\mathfrak{g}_T^{\mathrm{SC}} = \mathrm{Hom}_\Sigma(\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}}), \mathrm{End}_{(B_\partial, \mathcal{C}_{\mathrm{line}})})$. Its closed-colour projection $(m_k)_{k\geq 1}$ encodes the A_∞ chiral algebra on $\mathcal{A}$. Each $m_k\colon \mathcal{A}^{\otimes k} \to \mathcal{A}((\lambda_1))\cdots((\lambda_{k-1}))$ is a configuration-space integral on $\overline{C}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$, with the holomorphic factor producing OPE singularities and the topological factor producing the ordering. Summing over trees T with k leaves, $m_T(a_1, \ldots, a_k) = \int_{\overline{C}_k(\mathbb{C})} \omega_T \prod_v a_{(n_v)} b_v$. The Maurer–Cartan equation is the spectral Stasheff identity: Stokes' theorem on boundary strata of $\overline{C}_k(\mathbb{C})$ produces the A_∞ relations.

10.4. PVA descent (Theorem G)

On cohomology, the operations simplify. The regular part of m_2 is a commutative associative product $\mu(a,b) = \mathrm{Reg}_{z_1\to z_2} m_2(a(z_1), b(z_2))$; commutativity follows from the $\mathbb{Z}/2$ involution of $\overline{C}_2(\mathbb{C})$, associativity from the three FM_3 boundary strata. The singular part defines the lambda-bracket $\{a_\lambda b\} = \mathrm{Res}_{z_1\to z_2} e^{\lambda(z_1-z_2)} m_2(a(z_1), b(z_2))$; the exponential extracts all pole orders. All $m_{k\geq 3} = 0$ on cohomology because $\mathrm{Conf}_k(\mathbb{R})$ is contractible for $k \geq 3$.

The Arnold relation $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ on $\overline{C}_3(\mathbb{C})$ is equivalent to the three codim-1 boundary divisors of $\overline{M}_{0,4}$ summing to zero in homology; its residue against $a \otimes b \otimes c$ produces the Jacobi identity for $\{-_\lambda-\}$. The three-face Stokes identity applied to $\omega = \eta_{12} \wedge \eta_{23} \cdot a(z_1)b(z_2)c(z_3)$ produces the Leibniz rule $\{a_\lambda(b\cdot c)\} = \{a_\lambda b\}c + b\{a_\lambda c\}$. The exchange cylinder on $\overline{C}_2(\mathbb{C}) \times E_1(2)$ gives skew-symmetry $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$. Thus $(\mu, \{-_\lambda-\})$ is a (-1)-shifted Poisson vertex algebra, each axiom in one line from the geometry of FM.

(Vol I convention: we state the PVA in OPE modes; Vol II uses lambda-brackets throughout, and the two conventions are translated by the standard $\{a_\lambda b\} = \sum_{n\geq 0} \lambda^n a_{(n)} b/n!$.)

10.5. The Heisenberg algebra in Swiss-cheese coordinates

For $\mathcal{H}_k$: $m_2(J,J) = k\cdot\mathbf{1}$, $m_{k\geq 3} = 0$, $\{J_\lambda J\} = k\lambda$. All higher operations vanish because the OPE has no simple pole. The spectral R-matrix is the scalar $R(z) = \exp(k/z)$; YBE holds trivially. The PVA is the free bosonic PVA, the universal enveloping PVA of the rank-one abelian Lie conformal algebra.

10.6. The Koszul triangle

From C on $\widetilde{X}_D$ three independent objects arise:

(i) Boundary $A_b = \mathrm{End}_C(b)$: a chart.

(ii) Bulk $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}) = R\,\mathrm{Hom}_{\mathrm{Fun}(\mathcal{C},\mathcal{C})}(\mathrm{Id},\mathrm{Id})$: the universal bulk, Morita invariant, computed by chiral Hochschild cochains.

(iii) Lines $A^{!}_{b^{\infty}} = D_{\mathrm{Ran}}(\bar{B}(A_b))$: the homotopy Koszul dual, whose cohomology on the Koszul locus recovers the strict dual $A^{!}_{b} = (H^*(\bar{B}(A_b)))^{\vee}$. The module category $A^{!}_{b}$-mod with spectral braiding $R(z)$ is the braided tensor category of line operators.

The triangle of identifications

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}) \;\simeq\; \mathrm{HH}^{\bullet}(A_b) \;\simeq\; \mathrm{ChirHoch}^{\bullet}(A^{!}_{b}), \qquad A^{!}_{b}\text{-mod} = \{\text{line operators}\}$$

relates three distinct functors: endomorphisms (bulk), Ran-space Verdier duality (lines), and identity-bimodule endomorphisms (centre). The bar complex classifies twisting morphisms, not the bulk.

Spectral braiding. Stokes' theorem on $\overline{C}_3(\mathbb{C})$ applied to $\alpha \star \alpha \star \alpha$ decomposes the boundary into three pairwise collision divisors, one for each side of the Yang–Baxter equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$.

Bulk = derived centre. Let $\mathcal{B}_\partial$ be the boundary factorization algebra on $\mathbb{R}$. Its derived centre $Z_{\mathrm{der}}(\mathcal{B}_\partial) = R\,\mathrm{Hom}_{\mathcal{B}_\partial\text{-bimod}}(\mathcal{B}_\partial, \mathcal{B}_\partial)$ computes endomorphisms of the identity bimodule. The E_1-structure gives $Z_{\mathrm{der}}(\mathcal{B}_\partial) \simeq \mathrm{HH}^{\bullet}(\mathcal{B}_\partial)$ as E_2-algebra. The Swiss-cheese structure identifies the holomorphic bulk with this E_2-algebra.

10.7. Curved Swiss-cheese at $g \geq 1$

At genus $g \geq 1$, the bar complex carries curvature:

$$d^2_{\mathrm{fib}} \;=\; \kappa(\mathcal{A}) \cdot \omega_g,$$

with ω_g the Arakelov $(1,1)$-form representing the first Chern class of the relative dualising sheaf. The coproduct remains strictly coassociative: the interval $\mathbb{R}$ has no topology, so the curvature lives entirely in the holomorphic direction. The total corrected differential $D_g = d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}_g$ restores flatness by coupling the ordering to the Hodge bundle: $d^2_{\mathrm{fib}} + [d_{\mathrm{fib}}, \nabla^{\mathrm{GM}}] + (\nabla^{\mathrm{GM}})^2 = \kappa\omega_g - \kappa\omega_g = 0$.

The derived category must be replaced by the coderived category $\mathsf{D}^{\mathrm{co}}(\mathcal{A}^{(g)})$: ordinary $d^2 = 0$ fails at $g \geq 1$, every object is acyclic in the derived sense, and the correct homotopy theory comes from totalising the curvature bicomplex. Passage from derived to coderived is the categorified form of passage from $d^2 = 0$ to curved MC $d\alpha + \alpha^2 + \kappa\omega = 0$.

10.8. Lagrangian self-intersection

Volume II identifies the geometric substrate of the bar construction: the bar complex is the structure sheaf of the Lagrangian self-intersection $\mathfrak{S} = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$ in a (-2)-shifted symplectic stack. Every theorem is a face of that geometry: Theorem A is the groupoid comodule-module adjunction for $\mathfrak{S}$; Theorem B (bar-cobar inversion) reconstructs the Lagrangian from its clean self-intersection; Theorem C (complementarity) is two Lagrangians whose intersection carries a (-1)-shifted symplectic structure; Theorem D is the Kodaira–Spencer class; Theorem H (chiral Hochschild polynomial growth) is the HKR theorem for the Lagrangian embedding.

10.9. Factorization-primary hierarchy

$\mathrm{SC}^{\mathrm{ch,top}}$ is a *local model*: the formal completion of a factorisation structure at collision points. The global truth is a factorisable $\mathcal{D}$-module on $\mathrm{Ran}(\Sigma_g)$ over $\overline{\mathcal{M}}_g$. Six layers:

(i) Geometric substrate: smooth and stable nodal curves, Ran space, $\overline{\mathcal{M}}_{g,n}$.
(ii) Factorizable $\mathcal{D}$-modules (Beilinson–Drinfeld).
(iii) HT product geometry $\Sigma_g \times \mathbb{R}$.
(iv) Bar as factorization coalgebra on $\mathrm{Ran}(X)$.
(v) Koszul duality as factorization duality; Feynman involutivity $\mathrm{FT}^2 \simeq \mathrm{id}$.
(vi) Three models as three gauges: flat associated graded ($D_0^2 = 0$, operadic/local), corrected holomorphic ($D^{(g)2} = 0$, Fay trisecant, still local), curved geometric ($d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$, Arakelov, global).

The curvature measures what the local extraction loses: it is the monodromy of the $\mathcal{D}$-module connection around B-cycles of Σ_g. The modular bar complex $\bar{B}_{\mathrm{mod}}(\mathcal{A})$ is flat because it couples all genera simultaneously via D_1; the genus-fixed bar complex is curved because restriction decouples this connection.

The relative Feynman transform $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathrm{SC}^{\mathrm{ch,top}}}$ mediates between factorisation (global) and operadic (local) viewpoints. The modular bar coalgebra is an algebra over this transform; homotopy-involutivity plus factorisation Koszul duality gives the flat-derived / curved-coderived comparison at all genera.

10.10. Four-stage architecture

Stage 1. $\mathrm{SC}^{\mathrm{ch,top}}$ homotopy-Koszul; bar-cobar is Quillen-equivalent; chiral derived centre $\simeq C^\bullet_{\mathrm{ch}}(A_b, A_b)$ is the universal bulk. Proved.

Stage 2. Primitive object is the open factorisation dg-category $\mathcal{C}$; A_b is a chart; bulk is Morita invariant. Proved.

Stage 3. Tangential log curves (X, D, τ) provide bordered FM compactification; local-global bridge lifts operadic structure to factorisable $\mathcal{D}$-module on $\mathrm{Ran}(X)$. Proved locally; modest globally.

Stage 4. Trace and clutching on the open sector produce the full open/closed MC element Θ^{oc}. Annulus trace is proved; full chain-level clutching is programme.

Five negative principles: (N1) bar $\neq$ bulk; (N2) boundary algebra is a chart; (N3) tangential log geometry required; (N4) modularity is trace plus clutching on the open sector; (N5) dg-shifted Yangian is proved for affine lineage only, extension to non-standard families is programme.

11. Modular PVA quantization

The open/closed world supplies a genus-zero datum: the Swiss-cheese MC element α_T, the PVA $(\mu, \{-_\lambda-\})$ on cohomology, the KZ/BPZ shadow connections at degree n. All are tree-level. How does this structure extend across the genus filtration? The answer is a genus-by-genus obstruction theory whose input is the genus-0 PVA and whose output is the full modular MC element. The nonseparating degeneration operator D_1 is the primitive genus-raising differential; it squares to zero, anticommutes with the tree-level differential D_0, and defines a spectral sequence whose obstructions are the quantisation anomalies.

11.1. One-edge principle

Every nontrivial modular obstruction theory must contain a primitive genus-raising operator; otherwise any genus-0 MC element lifts tautologically to all genera. The one-edge expansion

$$D_{\mathrm{exp}} \;=\; D_{\mathrm{sep}} + D_{\mathrm{nsep}}, \qquad D_0 \;=\; D_{\mathrm{int}} + D_{\mathrm{sep}}, \qquad D_1 \;=\; D_{\mathrm{nsep}}$$

separates genus-preserving from genus-raising degenerations. $(D_0 + D_1)^2 = 0$ stratifies by genus shift: $D_0^2 = 0$, $D_0 D_1 + D_1 D_0 = 0$, $D_1^2 = 0$. Trees govern chirality; loops govern modularity. The three identities are codim-2 cancellation stratified by genus.

11.2. Modular bar datum

A modular bar datum on a reduced complete graded collection $F \mapsto C(F)$ consists of (i) an internal differential $d\colon C(F) \to C(F)$ with $d^2 = 0$; (ii) for every one-edge expansion $\epsilon \in \mathrm{Œ}(F)$ (separating or nonseparating), a degree-+1 map

$$\Delta_\epsilon\colon sC(F) \;\longrightarrow\; \overline{}(\epsilon) \otimes \bigotimes_{u \in V(\epsilon)} sC(\mathrm{Fl}_\epsilon(u));$$

(iii) the codim-2 cancellation $\sum_{\epsilon_1,\epsilon_2} \operatorname{sgn} \Delta_{\epsilon_2} \circ \Delta_{\epsilon_1} = 0$ for every two-edge graph; and (iv) $d \circ \Delta_\epsilon + \Delta_\epsilon \circ d = 0$. The total $D = d + \sum_\epsilon \Delta_\epsilon$ satisfies $D^2 = 0$ by three mechanisms: $d^2 = 0$, anticommutation, and the codim-2 axiom. This is the algebraic incarnation of the cell decomposition of $\overline{\mathcal{M}}_{g,n}$ into stable-graph strata.

11.3. Complete modular bar coalgebra

For a connected stable graph Γ of genus g, set

$$C[\Gamma] \;=\; \overline{}(\Gamma) \otimes \bigotimes_{v \in V(\Gamma)} sC(\mathrm{Fl}(v)), \qquad \overline{}(\Gamma) = \bigotimes_e \overline{}(e).$$

The complete modular bar coalgebra is

$$\mathbf{B}_{\mathrm{mod}}(C) \;=\; \prod_{[\Gamma] \in \mathsf{StGraph}^{\mathrm{conn}}} (C[\Gamma])_{\mathrm{Aut}(\Gamma)},$$

completed over total genus $g(\Gamma) = b_1 + \sum_v g(v)$. Edge contraction is the reduced coproduct; the coalgebra is conilpotent.

11.4. Strict modular deformation algebra

$L_{\mathrm{mod}}(C) := \mathrm{Coder}(\mathbf{B}_{\mathrm{mod}}(C))[-1]$ is complete filtered by genus, with convolution $L_{\mathrm{mod}}(C) \cong \mathrm{Hom}(\mathbf{B}_{\mathrm{mod}}(C), sC)[-1]$ under commutator bracket. The algebra is pro-nilpotent in F^1, and the associated graded inherits a dg Lie structure at each genus.

11.5. Genus-1 obstruction theorem

For a genus-0 MC element $\Theta_0 \in \mathrm{MC}(\mathrm{gr}^0_F L)$, the genus-1 obstruction is

$$\mathrm{Ob}_1(\Theta_0) = [D_1\Theta_0] \in H^2((\mathrm{gr}^1_F L)^{\Theta_0}, d_0^{\Theta_0}).$$

$D_1\Theta_0$ is a $d_0^{\Theta_0}$-cocycle by $D_0D_1 + D_1D_0 = 0$ and the MC equation for Θ_0. The class vanishes iff Θ_0 lifts to $\Theta_0 + \hbar\Theta_1$ satisfying the modular MC equation modulo $\hbar^2$.

In PVA coordinates, the nonseparating one-edge operator acts as the reduced odd Laplacian: for cyclic $(A, m_k, \langle -, - \rangle)$ with basis $\{e^i\}$ and dual basis $\{e_i\}$,

$$\ell_1^{(1)}(a) = \hbar\Delta_{\mathrm{cyc}}(a) = \hbar \sum_i m_2(e^i, m_2(a, e_i)).$$

The full Ob_1 is the cohomology class of $\Delta_{\mathrm{cyc}}(\Theta_0)$.

11.6. $\mathcal{W}_3$ computation

A filtered quadratic PVA is generated by $u^1, \dots, u^N$ with λ-bracket $\Pi^{ab}(\partial)$ of total generator degree ≤ 2. Classical $\mathcal{W}_3$ fits: generators T (weight 2), W (weight 3), with the composite $\Lambda =: TT : -(3/10)\partial^2 T$ controlling the nonlinear channel; the WW OPE has the $c = -22/5$ pole in $16/(22 + 5c)$ where Λ becomes null.

The triangular W-normal form vanishing theorem: in the local, translation-invariant, filtration-preserving sector, $\mathrm{Ob}_1(\Theta_0) = 0$ for classical $\mathcal{W}_3$. The relevant cohomology is $H^1((\mathrm{gr}^1 L)^{\Theta_0}, d_0^{\Theta_0}) \cong \mathbb{C} \cdot \partial_c P_c$, one-dimensional and spanned by the central-parameter direction. Every genus-1 lift is gauge equivalent to a renormalisation $c \mapsto c + \hbar c_1$; the genus-1 quantum $\mathcal{W}_3$ is the unique quantisation in the direction ∂_c.

11.7. Curved extension via S-tails

In the inhomogeneous quadratic setting (Gui–Li–Zeng), the correct input is a twisted pair (B, B°, S): a CDG-algebra B, a CDG-coalgebra B°, and curvature S. An S-tail expansion attaches a univalent S-decorated vertex to a corolla, followed by contraction of the new internal edge. Adjoining S-tail expansions to one-edge expansions with the same codim-2 compatibilities gives a curved modular bar object. The curved MC equation is $\mu((S + \alpha) \boxtimes (S + \alpha)) = 0$; at genus 0 it is the curved A_∞ MC equation with $m_0 = S$, at genus 1 it is $\Delta_{\mathrm{cyc}}(S + \alpha) = 0$, and at genus g it is the BV quantum master equation $\hbar\Delta(e^{(S+\alpha)/\hbar}) = 0$.

11.8. Genus-0 Gui–Li–Zeng comparison

At tree level, the modular MC equation reduces to the classical twisting morphism equation. For a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ with $\mathcal{A}^!$ a CDG algebra with curvature S,

$$\mathrm{MC}_0(L_{\mathrm{mod}}) \cong \mathrm{Hom}_{\mathsf{dg}}(\mathcal{A}, B^\circ) \hookrightarrow \mathrm{MC}(\mathcal{A}^! \widehat{\otimes} \mathcal{B}).$$

The modular theory is the genus extension of the classical Gui–Li–Zeng picture.

12. Holographic modular Koszul duality

The open/closed world (Section 10) supplies the 3d geometric framework; the modular PVA quantisation programme (Section 11) supplies the obstruction theory. The holographic modular Koszul datum packages both into a single algebraic object: a six-component structure that records the boundary algebra, its Koszul dual, the line-operator category, the spectral R-matrix, the universal MC element, and the modular shadow connection. Every ingredient constructed so far (the bar complex, the shadow tower, the Swiss-cheese structure, the PVA descent, the genus expansion) is a projection of this datum. The datum is the smallest package from which the full correspondence reconstructs.

12.1. The holographic modular Koszul datum

Every 3d HT system T is controlled by

$$\mathcal{H}(T) \;=\; (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}),$$

where:

(i) $\mathcal{A}$ is the boundary chiral algebra on a curve X, restricted from the holomorphic boundary $\{0\} \times X \subset \mathbb{R}_+ \times X$.

(ii) $\mathcal{A}^!$ is the strict chiral Koszul dual, $\mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee$ with $\mathcal{A}^{\mathrm{i}} = H^*(\bar{B}(\mathcal{A}))$. Holographically, $\mathcal{A}^!$ is a dg-shifted Yangian: the bar complex $\bar{B}(A_b)$ is an E_1 coassociative coalgebra, and $\mathcal{A}^!$ is its linear dual. The $\mathrm{SC}^{\mathrm{ch,top}}$ datum is the pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\ \mathcal{A})$ of chiral derived center and boundary algebra.

(iii) $\mathcal{C} \simeq \mathcal{A}^!$-mod is the braided tensor category of line operators: objects are topological defect lines, morphisms junction operators, braiding from crossing in the holomorphic plane.

(iv) $r(z) \in \mathcal{A}^! \otimes \mathcal{A}^!((z^{-1}))$ is the classical r-matrix satisfying CYBE, the binary genus-0 shadow $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$.

(v) $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ is the universal MC element from the bar-intrinsic construction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, always MC because $D^2_{\mathcal{A}} = 0$.

(vi) $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$ is the modular shadow connection on $\overline{\mathcal{M}}_{g,n}$, flat by the MC equation projected to (g, n).

12.2. Collision filtration

The modular convolution $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ carries a collision filtration $F^0_{\mathrm{coll}} \supset F^1_{\mathrm{coll}} \supset \cdots$ indexed by minimum pairwise distance on $\overline{C}_n(\mathbb{C})$. Depth 0 is free propagation; depth 1 is first collision; depth k is k-fold nested collision. The spectral sequence has $E_1^{p,*} = H^*(\mathrm{gr}^p_{\mathrm{coll}}\, \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, d_0)$ with d_1 Arnold cancellations; it converges to $H^*(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$, and collapse is governed by shadow depth: class $\mathbf{G}$ collapses at E_2; class $\mathbf{M}$ does not collapse at any finite page.

12.3. Four proved recovery theorems

(R1) Collision residue is twisting. At depth 1, degree 2: $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) = \pi_{\mathcal{A}}$, the universal twisting morphism producing the dg-shifted Yangian $r_{\mathcal{A}}(z)$. The MC equation becomes $\partial\pi + \pi \star \pi = 0$.

(R2) CYBE from Arnold. The three boundary divisors of $\overline{M}_{0,4}$ correspond to channels (12|34), (13|24), (14|23). Evaluating the Arnold relation on the Casimir $\Omega \otimes \Omega \in (\mathcal{A}^!)^{\otimes 3}$ produces the three terms of the CYBE $\sum [r_{ij}, r_{ik}] = 0$.

(R3) Affine shadow / KZ comparison. For $\mathcal{A} = \widehat{\mathfrak{g}}_k$, on the affine comparison surface the genus-0 degree-n shadow one-form is the KZ connection form

$$\mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) \;=\; \sum_{i<j} r_{ij}(z_i - z_j)\, d\log(z_i - z_j) \;=\; \frac{1}{h^\vee + k} \sum_{i<j} \frac{\Omega_{ij}\, dz_{ij}}{z_{ij}},$$

where $r_{ij} = \Omega_{ij}/(h^\vee + k)$ is the propagator-weighted Casimir. Flatness is MC at genus 0, degree n. For Virasoro, the Virasoro conformal Ward extraction plus higher collision residues yields the BPZ equations. The affine Drinfeld–Kohno theorem compares monodromy with the quantum-group representation category.

(R4) Quartic as L_∞ obstruction. $[\mathfrak{R}_4^{\mathrm{mod}}(\mathcal{A})] = [o_4(\mathcal{A})] \in H^2(\mathcal{A}^{\mathrm{sh}}_{4,0})$ is the obstruction to extending $r(z)$ from degree 2 to degree 4. Vanishes for classes **G**, **L**, **C**; nonzero for class **M** (Virasoro: $Q^{\mathrm{contact}} = 10/[c(5c+22)]$).

12.4. Five conjectural targets

- (i) *Boundary-defect realisation.* $\mathcal{A}_\partial(T)$ and $\mathcal{A}^!_\partial(T)$ are the two algebras of $\mathcal{H}(T)$ for every 3d HT system, with $\mathcal{C} \simeq \mathcal{A}^!$-mod.
- (ii) *Yangian-shadow.* In full derived generality, the collision residue recovers the dg-shifted Yangian $r(z)$ at the chain level, including higher A_∞ homotopies.
- (iii) *Sphere reconstruction.* The Gaiotto–Zhang (2026) commuting differentials on the non-planar sphere algebra are the scalar shadow $\mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})$; sphere amplitudes are flat sections of $\nabla^{\mathrm{hol}}_{0,n}$ on established comparison surfaces.
- (iv) *Quartic resonance obstruction.* $\mathfrak{R}_4^{\mathrm{mod}}$ is the first global nonlinear invariant, measuring the failure of the large-N expansion to truncate at tree level.
- (v) *Singular-fibre descent.* $\nabla^{\mathrm{hol}}|_{\partial\overline{\mathcal{M}}} \sim \nabla^{\mathrm{clutch}}$ with residues from the clutching law of $\Theta_{\mathcal{A}}$: $\Theta_{\mathcal{A}}|_{\sigma_\Gamma} = \prod_v \Theta_v \cdot \prod_e P_e$.

12.5. Coloured modular deformation object

A 3d HT theory produces three observation algebras: $\mathrm{Obs}^{\mathrm{hol}}$ (bulk), Obs^{∂} (boundary), Obs^{ℓ} (line). The coloured Swiss-cheese FM space $\overline{C}^{\log}_{\bullet,\mathrm{SC}}$ encodes the collision geometry of all three. The three-colour modular deformation object is

$$\mathfrak{g}^{\mathrm{HT,log}}_{\mathrm{mod}} \;=\; \hom^{\alpha_{\mathrm{HT}}}\big(C_\bullet(\overline{C}^{\log}_{\bullet,\mathrm{SC}}), \mathrm{End}_{\mathrm{Ch}_\infty}(\mathrm{Obs}^{\mathrm{hol}}_\infty, \mathrm{Obs}^{\partial}_\infty, \mathrm{Obs}^{\ell}_\infty)\big).$$

The three colours interact through $\mathrm{SC}^{\mathrm{ch,top}}$: bulk-to-boundary, boundary-to-line, no line-to-bulk.

The modular effective action is the graph sum

$$S^{\mathrm{mod}}_{\mathrm{HT}} \;=\; \sum_{\Gamma \in \mathsf{StGraph}_{\mathrm{SC}}} \frac{\hbar^{b_1(\Gamma)}}{|\mathrm{Aut}(\Gamma)|} W^{\log}_{\Gamma} O_\Gamma,$$

and the BV quantum master equation $(d_{\mathrm{BV}} + \hbar\Delta_{\mathrm{odd}})\exp(S^{\mathrm{mod}}/\hbar) = 0$ is the single equation governing T. At genus 0: classical BV $\{S_0, S_0\} = 0$; genus 1: one-loop anomaly cancellation; genus g: g-loop Ward identity.

12.6. Khan–Zeng bridge

Khan and Zeng construct a 3d HT Poisson sigma model on $\mathbb{R}_+ \times \mathbb{C}$ attached to a freely generated PVA: $\mathfrak{g}$-valued $(0,1)$-form $A_{\bar z}$, $\mathfrak{g}^*$-valued scalar ϕ, $\mathfrak{g}$-valued 1-form η on $\mathbb{R}_+$, with action

$$S_{\mathrm{KZ}} = \int_{\mathbb{R}_+\times\mathbb{C}} \left(\langle\phi, \bar\partial A\rangle + \langle\phi, \tfrac{1}{2}\{A_\lambda A\}|_{\lambda=0}\rangle + \langle\eta, d_t\phi\rangle\right) dt \wedge dz \wedge d\bar z.$$

Gauge invariance is the PVA Jacobi identity; a Virasoro element promotes HT to topological; the boundary carries a (-1)-shifted Poisson bracket and generates the genus-0 quantisation problem. The modular extension couples the bulk theory to stable curves at $g \geq 1$ and controls the full genus tower. The pipeline:

$$\text{classical coisson} \xrightarrow{\text{KZ quantisation}} \text{genus-0 quantum Koszul} \xrightarrow{\text{modular lift}} \mathcal{H}(T).$$

The bridge is completed in Volume II by the doubling theorem: the half-space BV problem for any logarithmic $\mathrm{SC}^{\mathrm{ch,top}}$-algebra embeds into the doubled whole-space via method of images; the reflected weight identity is the σ-invariant restriction of the standard Stokes identity on the doubled FM compactification. The affine Poisson–vertex sigma model satisfies the physics bridge hypotheses, and Drinfeld–Sokolov reduction preserves this structure. The entire standard landscape enters through these two gates.

The genus expansion of the modular bar differential, $D = \sum_{g\geq 0}\hbar^g D^{(g)}$, is the Feynman transform of the modular operad; the canonical MC element $\delta_{\mathcal{A}} = D - D^{(0)}$ computes positive-genus corrections. Stable graphs, automorphism weights, genus bookkeeping: the mathematical shadow of the 't Hooft expansion. The identification $\hbar \leftrightarrow 1/N$ for a specific holographic model remains the principal open bridge.

12.7. Heisenberg holographic datum

Base case: shadow class **G**, shadow depth 2, trivial line-operator category, scalar R-matrix. For $\mathcal{A} = \mathcal{H}_k$:

$$\begin{aligned} &\mathcal{A} = \mathcal{H}_k, && \mathcal{A}^! = \mathrm{Sym}^{\mathrm{ch}}(V^*), && C = \mathsf{Vect}, \\ &r(z) = k/z, && R(z) = \exp(k/z), && \Theta_{\mathcal{H}_k} = k\cdot\eta\otimes\Lambda. \end{aligned}$$

The shadow connection $\nabla^{\mathrm{hol}}_{g,n} = d$ is trivial (scalar MC element). All five conjectural targets are trivially verified: free boundary, curved symmetric dual (with $\mathcal{H}_k^! \neq \mathcal{H}_{-k}$, $m_0 = -k\omega$), trivial lines, vanishing quartic, no moduli dependence.

12.8. Affine $\widehat{\mathfrak{sl}}_2$ holographic datum

First nontrivial example: class **L**, shadow depth 3, Yang's $r(z) = k\,\Omega/z$, line-operator category $Y(\mathfrak{sl}_2)$-mod, KZ shadow connection, quantum-group representation theory from genus-0 monodromy.

$$\mathcal{H}(\widehat{\mathfrak{sl}}_{2k}) \;=\; \Big(\widehat{\mathfrak{sl}}_{2k},\; \widehat{\mathfrak{sl}}_{2-k-4},\; \mathcal{O}_q^{\mathrm{sh}},\; \tfrac{k\,\Omega}{z},\; \Theta_k,\; d-\kappa_k\,\omega_g\Big),$$

with $q = e^{i\pi/(k+2)}$ and $\kappa_k = 3(k+2)/4$. At $k=-2$: $\kappa = 0$, the Feigin–Frenkel centre is the algebra of $\mathfrak{sl}_2$-opers. The KZ monodromy is compared with the $U_q(\mathfrak{sl}_2)$ R-matrix by affine Drinfeld–Kohno; on the reduced evaluation comparison surface, $\mathcal{C}^{\mathrm{red}}_{\mathrm{line}}|_{\mathrm{eval}}$ is identified with the corresponding quantum-group side, and the coloured Jones polynomial is recovered from monodromy of the bar complex (Steinberg convolution of the chiral self-intersection on $\overline{C}_k(\mathbb{C})$).

This is 3d Chern–Simons at level k. The holographic datum is the $k=0$-vanishing r-matrix plus the full KZ package.

12.9. Virasoro holographic datum

First example of shadow class **M**. The r-matrix $r_c(z) = (c/2)/z^3 + 2T/z$ has both a third-order pole (gravitational) and a first-order pole (stress-tensor exchange); the quartic contact invariant $Q^{\mathrm{contact}} = 10/[c(5c+22)] \neq 0$. The datum packages 3d gravity on $\mathbb{R}_+ \times \mathbb{C}$:

$$\begin{aligned}
\mathcal{H}(\mathrm{Vir}_c) &= (\mathrm{Vir}_c,\; \mathrm{Vir}_{26-c},\; \mathcal{C}_c,\; r_c(z),\; \Theta_c,\; \nabla_c^{\mathrm{hol}}),\\
r_c(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, \qquad \kappa_c \;=\; \frac{c}{2},\\
\nabla^{\mathrm{hol}}_{1,1} &= d \;-\; \kappa_c\,\omega_1 \;-\; \frac{10}{c(5c+22)}\,\delta_4 \;-\; \cdots.
\end{aligned}$$

Koszul dual Vir_{26-c} (self-dual at $c=13$, not $c=26$); shadow connection with curvature $\kappa_c\omega_g$ at leading order. The transferred coproduct Δ_z^{Vir} is strictly primitive at all degrees (Principle **??**): ghost-number obstruction intrinsic to DS reduction kills every HPL tree correction. All gravitational dynamics resides in the collision residue and the infinite A_∞ product tower; the coproduct channel is identically zero. Gravity does not produce particles, only braids them.

12.10. The M2 brane example

First genuinely interacting holographic example: boundary algebra $\mathcal{W}_{1+\infty}$ (infinitely many generators), Koszul dual dg-shifted Yangian for $\mathfrak{psl}(2|2)$, line-operator category a quantum affine superalgebra. All three genus-2 shells active; the planted-forest shell receives its first nontrivial contribution from Mok's log-FM strata.

Physical setup: 3d $\mathcal{N}=4$ ADHM gauge theory, N M2 branes probing $\mathbb{C}^2/\mathbb{Z}_k$. At large N, the boundary VOA is $\mathcal{W}_{1+\infty}[c]$ at $c \sim N$, with strong generators of weights $1, 2, 3, \ldots$ and rational OPE coefficients in N.

The Koszul dual is $\mathcal{A}^!_{M2,\infty} \simeq Y_\hbar^{\mathrm{dg}}(\mathfrak{psl}(2|2))$. The classical r-matrix $k\,\Omega_{\mathfrak{psl}(2|2)}/z$ vanishes at $k=0$ and satisfies CYBE via the Arnold relation on the $\mathfrak{psl}(2|2)$ Casimir. The line-operator category is $\mathcal{C}_{M2} \simeq \mathrm{Rep}(U_q(\widehat{\mathfrak{psl}}(2|2)))$ with $q = e^{2\pi i/(k+2)}$.

The datum:

$$
\begin{aligned}
\mathcal{A} &= \mathcal{A}_{M2,\infty}, & \mathcal{A}^! &= Y_\hbar^{\mathrm{dg}}(\mathfrak{psl}(2|2)),\\
\mathcal{C} &= \mathrm{Rep}(U_q(\widehat{\mathfrak{psl}}(2|2))), & r(z) &= k\,\Omega_{\mathfrak{psl}(2|2)}/z,\\
\Theta_{M2} &= \sum_\Gamma \frac{W_\Gamma^{\log\overline{C}}}{|\operatorname{Aut}(\Gamma)|}\otimes\Phi_\Gamma^{M2}, & \nabla_{g,n}^{\mathrm{hol}} &= d - \mathrm{Sh}_{g,n}(\Theta_{M2}).
\end{aligned}
$$

Shadow tower: cubic nonzero (nonabelian current subalgebra); quartic nonzero ($16/(22+5c(N))$); quintic $\{\mathfrak{C}_{M2}, Q^{\mathrm{contact}}_{M2}\}_H \neq 0$; tower infinite. At genus 2, all three shells active; the planted-forest shell is the first contribution from Mok's log-FM strata to the holographic genus expansion. The quartic contact invariant is the first obstruction to planarity: it distinguishes the full string-theoretic amplitude from its planar limit. The modular Koszul datum $\mathcal{H}(M2)$ packages the complete holographic correspondence into a single MC problem.

Summary. Sections 1–2 constructed the bar complex from $\eta = d\log(z_1 - z_2)$ and the Arnold relation. Sections 3–4 identified these as projections of $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. Sections 5–6 extracted the characteristic invariants from $\Theta_{\mathcal{A}}$ and classified algebras by shadow depth. Section 7 ran the machine on every standard family, reading both the geometric face and the E_1 algebraic face for each family. Section 8 lifted the shadow obstruction tower into number theory. Section 9 proved scalar saturation and the Koszulness meta-theorem. Sections 10–11 transported the machine to three dimensions via the Swiss-cheese operad and the modular PVA quantisation obstruction theory. Section 12 packaged the full holographic correspondence into a six-component modular Koszul datum controlled by a single MC equation.

13. Completion and the frontier

The holographic datum of Section 12 packages a chiral algebra, its Koszul dual, the line-operator category, the spectral R-matrix, the universal MC element, and the shadow connection into a single algebraic object. The frontier consists of structural questions the packaging forces: the completed bar-cobar problem by infinite-type algebras, the analytic sewing problem by $g \geq 1$ curvature, the factorization envelope problem by the passage from Lie conformal input to modular output, and the non-principal duality problem by Drinfeld–Sokolov reduction at arbitrary nilpotent orbits.

13.1. The Maurer–Cartan frontier

Status: master conjectures MC1 through MC4 are proved; MC5 is partially proved, with the analytic HS-sewing package established at all genera and the genuswise BV/BRST/bar identification conjectural. MC3 holds for all simple types on the evaluation-generated core via multiplicity-free ℓ-weights (Theorem ??, Corollary ??). The residual problem DK-4/5 (extension beyond evaluation modules) is downstream.

MC1 (PBW concentration). For every standard family (Kac–Moody, Virasoro, principal $\mathcal{W}_N$, $\beta\gamma$, lattice) and every genus $g \geq 1$, the PBW spectral sequence on the genus-g bar complex has $E_\infty^{p,q}(g) = 0$ for $q \neq 0$. Proof: genus-0 Koszulness gives concentration of $E_1^{p,q}(g{=}0)$; the genus-g enrichment from the Hodge bundle is d_0^{PBW}-exact because it arises from the regular Eisenstein series (exact under Poincaré residue). Concentration is uniform in k.

MC2 (universal $\Theta_{\mathcal{A}}$). The bar-intrinsic construction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is automatically MC because $D^2_{\mathcal{A}} = 0$. Scalar saturation universality: for every one-parameter algebraic family $\mathcal{A}_k$ with rational OPE coefficients, $\Theta_{\mathcal{A}_k} = \kappa(\mathcal{A}_k) \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}}$. Proof via Whitehead reduction ($E_2^{p,q} = H^p(\mathfrak{g}; \mathrm{Sym}^q V) = 0$ at $p = 2$ by Whitehead's lemma for reductive $\mathfrak{g}$) and algebraic semicontinuity (the non-vanishing locus is Zariski-closed, proper, and empty at the free-field point). Ambient $D^2 = 0$ on the planted-forest complex follows from Mok's normal-crossings result.

MC3 (thick generation). On the type-A Yangian module surface, the four-package MC3 problem reduces to a single compact/completed extension packet via (a) chromatic filtration of K_0 by q-character support, (b) prefundamental Clebsch–Gordan closure $V_n \otimes L^- = \bigoplus_{j=0}^{n} L^-(n-2j)$ at character level, (c) Francis–Gaitsgory pro-nilpotent completion, and (d) the remaining DK compact/completion packet. The CG decomposition is extended to all simple types via the Chari–Moura multiplicity-free ℓ-weight property, replacing the minuscule hypothesis; the remaining post-CG completion packets beyond type A are the downstream problem.

MC4 (completed bar-cobar). The strong completion-tower theorem: the strong filtration axiom $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ yields an automatic degree cutoff, so for fixed N only finitely many degrees contribute to F^N, making continuity and Mittag-Leffler automatic. The completion closure $\mathrm{CompCl}(\mathcal{F}_{\mathrm{ft}})$ of complete filtered objects with finite-type associated graded carries a quasi-inverse bar-cobar homotopy equivalence. Key results: coefficient stability, MC twisting closure, completed twisting representability, and the uniform PBW bridge (MC1 implies MC4 for all standard families).

MC4 splits into two sub-problems, both proved:

- *MC4*$^+$ (positive towers: $\mathcal{W}_{1+\infty}$, affine Yangians, RTT algebras): solved by weight stabilisation.
- *MC4*^{0} (resonant towers: Virasoro, non-quadratic $\mathcal{W}_N$): reduced to a finite resonance problem by the resonance-filtered bar-cobar theorem. The resonance rank $\rho(\mathcal{A})$ measures the depth; every positive-energy chiral algebra has $\rho < \infty$ (Theorem ??). The Virasoro case: Vir_{26-c} is the depth-zero resonance shadow; the genuine $\mathcal{W}_\infty$-type dual requires the full resonance-filtered completion.

MC5 (genus expansion). Partially proved. The analytic sewing package is established at all genera, together with the algebraic genus tower. The full genuswise BV/BRST/bar identification remains conjectural. At genus 0, the algebraic BRST/bar comparison is proved (Theorem ??); the tree-level amplitude pairing is conditional on Corollary ??; bar-cobar inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is Theorem B. At genus 1, the curved bar complex with $d^2_{\mathrm{fib}} = \kappa \cdot \omega_1$ supplies the one-loop package, and the complementarity splitting $Q_1(\mathcal{A}) \oplus Q_1(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_{1,n}, Z(\mathcal{A}))$ is the genus-1 modular-invariance constraint.

At $g \geq 2$ the resolution proceeds in five steps:

(i) *Inductive genus determination* (Theorem ??): $\Theta^{(g)}_{\mathcal{A}}$ is determined by $\Theta^{(<g)}_{\mathcal{A}}$ via the graph-sum recursion on $\overline{\mathcal{M}}_{g,n}$.

(ii) *2D convergence*: no UV renormalisation on curves; the Feynman integrals $W_\Gamma = \int_{\sigma_\Gamma} \omega_\Gamma$ converge absolutely on each stratum without counterterm subtraction.

(iii) *Analytic-algebraic comparison* $D^{\mathrm{an}} = D^{\mathrm{alg}}$ on the algebraic core.

(iv) *General HS-sewing* (Theorem ??): polynomial OPE growth plus subexponential sector growth implies the Hilbert–Schmidt sewing condition, yielding trace-class closed amplitudes at every genus.

(v) *Heisenberg one-particle sewing* identifies the Heisenberg sewing amplitude with a Fredholm determinant, supplying the decisive first-case verification.

At multi-weight $g \geq 2$ (all-weight), the genus expansion admits an explicit cross-channel correction: $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$, proved for $\mathcal{W}_3$ by graph-by-graph computation over all seven stable graphs at $g = 2$ and all forty-two stable graphs at $g = 3$. The scalar formula of the uniform-weight expansion *fails* for multi-weight algebras at $g \geq 2$ (resolving Open Problem **??** negatively). A universal closed form for the gravitational cross-channel correction covers principal $\mathcal{W}_N$: $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = (N-2)(N+3)/96 + (N-2)(3N^3 + 14N^2 + 22N + 33)/(24c)$, exact for $\mathcal{W}_3$ and a lower bound for $N \geq 4$. The correct receptacle at $g \geq 1$ remains the coderived category: curvature forces coderived/contraderived passage.

Completion programme: q-windows and completion entropy. The bar-cobar quasi-isomorphism $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is proved at each finite-type stage $\mathcal{W}_N$. The passage to the inverse limit $\mathcal{W}_\infty = \varprojlim \mathcal{W}_N$ requires completed bar-cobar; the strong filtration axiom gives a quotientwise-finite differential. The completed bar complex decomposes into finite windows indexed by reduced weight $\rho(a) = \mathrm{wt}(a) - 1$.

For Virasoro at $q = 2$: the window $K_2(\mathrm{Vir})$ has basis $\{sT, sT|sT\}$, and the bar differential on the two-dimensional complex is $d(sT|sT) = s(T \cdot T)$. The product $T \cdot T$ decomposes into the quasi-primary Λ (weight 4, reduced weight $3 > 2$, projects to zero), $\partial^2 T$ (reduced weight $3 > 2$, projects to zero), and the central term $(c/2) \cdot \mathbf{1}$. Only the central term survives:

$$d(sT|sT)|_{K_2} = (c/2) \cdot s\mathbf{1}.$$

The modular characteristic $\kappa(\mathrm{Vir}_c) = c/2$ is the sole surviving coefficient of the bar differential on the smallest nontrivial window.

Window dimensions grow exponentially. Define the completion entropy $h_K(\mathcal{A}) := \log(\rho_{\mathcal{A}}^{-1})$, with $\rho_{\mathcal{A}}$ the radius of convergence of the primitive generating series from the vacuum character. The entropy measures kinematic completion complexity and forms a monotone ladder

$$h_K(\mathcal{W}_2) \approx 0.567 < h_K(\mathcal{W}_3) \approx 0.772 < h_K(\mathcal{W}_4) \approx 0.835 < \cdots < h_K(\mathcal{W}_\infty) \approx 0.872,$$

governed at the limit by the MacMahon plane-partition generating function $\chi_{\mathcal{W}_\infty}(q) = \prod_{n \geq 2}(1 - q^n)^{-(n-1)}$.

13.2. The analytic frontier

Curvature forces this direction. At $g \geq 1$ the bar complex satisfies $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g \neq 0$; the ordinary derived category is empty, and the correct receptacle is the coderived category, requiring Hilbert-space-valued structures.

The bare VOA is a dense algebraic skeleton; the object for partition functions is the sewing envelope $\mathcal{A}^{\mathrm{sew}}$, the Hausdorff completion in the topology of all-amplitude seminorms $\|a\|_{\mathrm{amp}} := \sup_{v \in \mathcal{V}} \langle v, \pi(a) v \rangle$. The HS-sewing condition:

$$\sum_{a,b,c \geq 0} q^{a+b+c} \|m_{a,b}^c\|_{\mathrm{HS}}^2 < \infty$$

for $|q| < 1$ implies closed amplitudes $\mathrm{Tr}(\pi(a) \cdot q^{L_0})$ are trace-class and the sewing operation converges absolutely. An analytic Koszul pair is one for which the analytic bar-cobar adjunction restricts to a Quillen equivalence on sewing-complete categories.

Four targets (two proved, two conjectural):

(a) *Heisenberg sewing* (decisive first case, proved): sewing envelope of $\mathcal{H}_k$ is $\mathrm{Sym}\, A_2(D) \cong$ ind-Hilb$(\mathcal{H}_k)$ (Moriwaki 2026).
(b) *Lattice sewing* (proved): sewing envelope of V_Λ is vertex-algebraic completion by lattice theta functions.
(c) *Analytic realisation criterion* (conjectural): HS-sewing on $\mathcal{A}$ plus dual HS on $\mathcal{A}^!$ should suffice.
(d) *Boundary bar duality* (conjectural): $B^{\mathrm{an}}(\mathcal{A}_\partial) \simeq B^{\mathrm{top}}(\mathrm{Fact}_T^{\mathrm{bulk}})$.

13.3. Factorization envelopes and the modular Koszul datum

The bar construction starts from a chiral algebra, but the natural input in most geometric constructions is a Lie conformal algebra. The factorisation envelope is the functor producing a chiral algebra from Lie conformal data; the programme asks whether the entire modular Koszul package can be constructed functorially at the Lie conformal level, without passing through the vertex algebra. The target is a universal modular factorization envelope $U_X^{\mathrm{mod}}(L)$, left adjoint to extracting primitive currents.

A cyclically admissible Lie conformal algebra L (conformal grading, compatible decreasing filtration, bounded poles, invariant bilinear pairing) produces the modular Koszul datum

$$\Pi_X(L) \;=\; \big(\mathrm{Fact}_X(L),\ \bar{B}_X(L),\ \Theta_L,\ \mathcal{L}_L,\ (V_L^{\mathrm{br}}, T_L^{\mathrm{br}}),\ R_4^{\mathrm{mod}}(L)\big),$$

with each component functorial in L. The target universal envelope $U_X^{\mathrm{mod}}(L)$ satisfies the adjunction

$$\mathrm{Hom}_{\mathsf{Fact}^{\mathrm{mod}}}(U_X^{\mathrm{mod}}(L), \mathcal{F}) \;\cong\; \mathrm{Hom}_{\mathsf{LCA}^{\mathrm{cyc}}}(L, \mathrm{Prim}^{\mathrm{mod}}(\mathcal{F})),$$

and $\Theta_L = \Theta^{\mathrm{env}}_{\le\infty}(L)$ is the universal MC class. Direct sums: κ, T^{br}, Δ, R_4^{mod} decompose additively. BRST reduction: $\Theta_{\mathcal{W}_N} = H^\circ_{Q_{\mathrm{DS}}}(\Theta_{V_k(\mathfrak{g})\otimes F_{\mathrm{gh}}})$, the shadow tower of the W-algebra is the DS reduction of the full current-plus-ghost tower, not of the bare affine tower.

13.4. Non-principal W-algebra duality

Drinfeld–Sokolov reduction connects affine landscape to W-algebra landscape. For principal f, the reduction commutes with bar-cobar duality by Feigin–Frenkel, and the holographic datum descends. For non-principal f, this commutation is unproved. The theory demands the extension: non-principal $\mathcal{W}$-algebras include the physically central cases ($\mathcal{W}_k(\mathfrak{sl}_N, f_{\mathrm{sub}})$ controls $\mathcal{N} = 2$ superconformal theories) and the mathematically central ones (Feigin–Tipunin at admissible levels).

The hook-type nilpotent orbit in type A is the first non-principal corridor, conditional on DS-bar compatibility (Theorem ??). For a partition $\lambda = (n - p, 1^p)$ in $\mathfrak{sl}_N$,

$$(\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda))^! \;\simeq\; \mathcal{W}_{k_\lambda^\vee}(\mathfrak{sl}_N, f_{\lambda^t}),$$

where λ^t is the transpose and $k_\lambda^\vee$ is a rational function of k. Proof: for hook-type partitions, DS commutes with bar-cobar by Fehily and Creutzig–Linshaw–Nakatsuka–Sato (2023–2025).

The transport propagation lemma: hook-type seeds combined with edge-compatibility propagate duality results along edges of the nilpotent Hasse diagram in type A. The type-A transport-to-transpose conjecture: for every partition λ of N, the duality formula holds. The general obstruction is proving that DS reduction commutes with bar-cobar at the chain level, i.e. that the natural map $\mathrm{DS}(\bar{B}(\widehat{\mathfrak{g}}_k)) \to \bar{B}(\mathrm{DS}(\widehat{\mathfrak{g}}_k))$ is a quasi-isomorphism.

13.5. The boxed machine

$$\boxed{\text{homotopy chiral input } \mathcal{A} \;+\; \text{log-FM cutting cooperad } C^{!}_{\mathrm{ch}} \;+\; \text{modular convolution } L_\infty \;\Longrightarrow\; \Theta_{\mathcal{A}},\ \mathcal{T}^{\mathrm{mod}}_{\Theta_{\mathcal{A}}},\ \kappa,\ \Delta_{\mathcal{A}},\ \mathfrak{R}^{\mathrm{mo}}_{4}}$$

The modular convolution L_∞-algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the single organising structure. It carries a natural modular L_∞ structure from the Feynman transform of the modular operad; $\mathrm{Conv}_{\mathrm{str}}(C^{!}_{\mathrm{ch}}, \mathcal{P}^{\mathrm{ch}})$ is its strict model. The universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ is proved: it is the bar-intrinsic extraction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, automatically MC because $D^2_{\mathcal{A}} = 0$.

The shadow obstruction tower consists of finite-order projections:

$$\begin{array}{lll}
\Theta^{\leq 2}_{\mathcal{A}} : & \kappa(\mathcal{A}) & \text{(modular characteristic, scalar curvature),} \\
\Theta^{\leq 3}_{\mathcal{A}} : & \mathfrak{C}(\mathcal{A}) & \text{(cubic shadow, Lie structure),} \\
\Theta^{\leq 4}_{\mathcal{A}} : & \mathfrak{Q}(\mathcal{A}) & \text{(quartic resonance class, contact),} \\
\Theta^{\leq r}_{\mathcal{A}} : & \mathrm{Sh}_r(\mathcal{A}) & \text{(degree-}r\text{ shadow),} \\
\Theta_{\mathcal{A}} : & \varprojlim_r \Theta^{\leq r}_{\mathcal{A}} & \text{(universal MC element).}
\end{array}$$

Theorems A–D and Theorem H are proved projections at the scalar level κ. The holographic datum $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, C, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ packages $\Theta_{\mathcal{A}}$ and its projections into the data of a 3d HT field theory.

Principle 0.0.1 (The four-test interface). *The modular Koszul machine has a complete algebraic-geometric interface with $\overline{\mathcal{M}}_{g,n}$ consisting of four independent proved tests:*

(i) $D^2_{\mathcal{A}} = 0$*: the bar differential squares to zero at all genera and degrees (Theorems **??**, **??**).*

(ii) $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ *for uniform-weight algebras at all genera , and unconditionally at genus 1 for all families; for multi-weight algebras at $g \geq 2$ (all-weight), with cross-channel correction $\delta F^{\mathrm{cross}}_g$ (Theorem **??**, Theorem **??**).*

(iii) $Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$*: complementarity assembles the Koszul pair into a Lagrangian decomposition (Theorem **??**).*

(iv) *Sewing: genus-g amplitudes converge from lower-genus data (Theorem **??**).*

*Each test captures a distinct aspect: (i) is purely algebraic and requires no Koszulness; (ii) identifies the tautological class but does not involve $\mathcal{A}^!$; (iii) requires the Koszul pair but is genus-by-genus; (iv) is analytic and independent of the Hodge-class identification. Any three can hold with the fourth failing. Three regions lie outside: multi-weight at $g \geq 2$ (all-weight, resolved with cross-channel correction, Open Problem **??**), BV/BRST = bar at higher genus (Conjecture **??**), and non-perturbative completion.*

The holographic frontier. The holographic datum $\mathcal{H}(T)$ is a boundary object. Its lift to a genuine 3d bulk theory is controlled by the relative holographic deformation complex $\mathfrak{h}^{\Theta}_{T} := \mathrm{fib}(\mathfrak{g}^{\mathrm{SC}}_{T} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{\alpha_T}$: lifts of $\Theta_{\mathcal{A}}$ are MC elements of the twisted fibre. The graded obstruction tower $[o_r(\beta)] \in H^2(\mathrm{gr}^F_r\, \mathfrak{h}^{\Theta}_{T})$ turns the frontier into cohomology calculus. At depth 4, the relative quartic obstruction $\mathfrak{R}^{\mathrm{rel}}_4(T)$ is the first invariant detecting when a boundary datum cannot arise from a planar holographic realisation. The Loop–Connes principle $\beta^*_{\mathrm{der}}(\Delta_{\mathrm{open}}) = \Lambda_P$: closed genus creation is the image of the open BV operator under the derived-centre map; loops in the bulk are traces on the boundary.

Cyclic trace, ribbon structure, and $1/N$. A bare modular graph sum does not define a literal 't Hooft expansion (Lemma **??**). Cyclic trace structure on the open E_1-chiral algebra resolves this: cyclicity enables ribbon thickening (Theorem ??), and a matrix realisation gives the exact weight $N^{\chi(\Sigma_\Gamma)}$ (Theorem ??). The three levels of language (plain modular E_1 genus expansion, cyclic traced modular E_1 't Hooft expansion, ribbonised modular Swiss-cheese holographic 't Hooft language) are made precise in the E_1 modular Koszul chapter (Chapter ??).

13.6. Admissible levels and logarithmic CFT

At admissible levels $k = -h^\vee + p/q$ (coprime $p \geq h^\vee, q \geq 1$), the Zhu algebra $A(\mathcal{V}_k(\mathfrak{g}))$ is semisimple but the vertex algebra is not simple: there is a maximal proper ideal J_k with $L_k(\mathfrak{g}) = \mathcal{V}_k(\mathfrak{g})/J_k$. The bar complex of $\mathcal{V}_k$ acquires a periodic CDG structure indexed by q-th roots of unity: the curvature $m_0 = \kappa \cdot \omega_g$ factors through $\Phi_q(\zeta)$, and the weight filtration has a finite plateau at level $N = q$. This is the algebraic door to logarithmic CFT on the degenerate admissible lane. On the non-degenerate admissible lane the simple quotient is rational; on the degenerate lane the simple quotient has a logarithmic module category. Weightwise finite-page character degeneration holds on the non-degenerate lane; identification of the resulting bar cohomology with logarithmic extensions is not proved here.

13.7. Relative Feynman transform

The modular bar coalgebra $\bar{B}_{\mathrm{mod}}(\mathcal{A})$ is an algebra over the relative Feynman transform $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathrm{SC}^{\mathrm{ch,top}}}$, the algebraic skeleton shared by the factorisation approach (§10.8) and the operadic approach. Factorisation on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$ over $\overline{\mathcal{M}}_g$ produces it via families of factorisable $\mathcal{D}$-modules; the operadic structure on $\mathrm{SC}^{\mathrm{ch,top}}_{\mathrm{mod}}$ produces it via formal completion.

Recognition. The tree-level part D_0 is the E_1 coassociative coalgebra structure (encoding boundary A_∞ data); the genus-raising part D_1 is the $\mathsf{Com}_{\mathrm{mod}}$-modular structure. The $\mathrm{SC}^{\mathrm{ch,top}}$ datum is the pair $\big(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A}\big)$ of chiral derived center and boundary algebra.

Homotopy-involutivity. $\mathrm{FT}^2_{\mathsf{Com}_{\mathrm{mod}}/\mathrm{SC}^{\mathrm{ch,top}}} \simeq \mathrm{id}$. Genus-0 involutivity from operadic homotopy-Koszulity; closed-colour modular involutivity from the Feynman involution (Theorem ??); assembly via the genus-filtration spectral sequence.

Three routes unified. Route B (factorisation) is global truth; Route A (operadic) is local approximation; Route C (this section) is the categorical image. The factorisation structure projects to the relative Feynman transform by forgetting $\mathcal{D}$-module data; the operadic structure injects by formal completion.

13.8. Computational infrastructure

Every formula has been verified by an independent symbolic computation layer comprising tens of thousands of tests across several hundred modules, covering the shadow obstruction tower through degree 32, the standard landscape for all seven families, the genus-2 and genus-3 planted-forest formulas, the DS-shadow cascade (degrees 2–8 for $N = 2, \ldots, 5$), the $\mathcal{W}_3$ multi-generator shadow metric, the Böcherer bridge at genus 2, the Pixton shadow bridge, the entanglement programme, the non-simply-laced bar computations (B_2, C_2, G_2, F_4), minimal model bar and fusion rules, the dressed propagator engine (scalar complementarity, R-matrix $R_1 = 1/12$), and the constrained Epstein zeta. The computation is adversarial: tests encode cross-family consistency checks that cannot be fooled by hardcoded values.

Volume III extends the engine to Calabi–Yau geometry: forty-two CY/BKM compute engines with 5700+ tests cover the $K3 \times E$ programme (shadow-Siegel gap, BKM root multiplicities, moonshine multiplier, Schottky barrier), toric CY_3 CoHA integrality, and the automorphic correction as shadow obstruction tower identification.

Volume II develops the 3d theory in three new chapters: the factorisation Swiss-cheese algebra (global truth on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$), the modular Swiss-cheese operad (local approximation), and the relative Feynman transform (algebraic skeleton). Together with the 3d gravity chapter, the affine half-space BV theorem, and eleven additional computational engines, these constitute the Volume II programme.

13.9. The single open problem

Everything proved in this monograph is a finite-order projection of the universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$. Everything open is a question about its ambient:

$$\boxed{\textit{Characterise the modular convolution algebra } \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \textit{ as a geometric object.}}$$

Precisely: the modular convolution algebra is the endomorphism algebra of the Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\mathrm{vac}}(\mathcal{A})$ in the (-2)-shifted symplectic stack of vacua. Volume I computes the Taylor jets of this embedding (the shadow obstruction tower). Volume II identifies the local composition law (the Swiss-cheese operad). What remains is the global geometry of the stack itself: the derived symplectic structure, the moduli of Lagrangians, and the higher-categorical descent that would make the construction independent of choices.

The approximately 150 open conjectures in these volumes are projections of this single question:

- the $\mathcal{D}$-module purity conjecture asks whether the Lagrangian is pure (characterisation (xii) of Koszulness);
- the non-principal $\mathcal{W}$-duality programme asks how the Lagrangian transforms under DS reduction at arbitrary nilpotent orbits;
- the analytic completion programme asks when the algebraic Lagrangian extends to a convergent one (the sewing envelope);
- the factorization envelope programme asks for the universal construction of the Lagrangian from Lie conformal data alone;
- the holographic completion question asks whether the genus-0 Lagrangian extends over $\overline{\mathcal{M}}_{g,n}$;
- the Calabi–Yau quantum group programme asks whether the automorphic correction of a generalised Borcherds–Kac–Moody superalgebra is the shadow obstruction tower of an E_2-chiral algebra constructed from a CY category (proved for $d = 2$; conditional on chain-level $\mathbb{S}^3$-framing for $d = 3$; Volume III).

Each is a different facet of the geometry of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$, and none requires going beyond the framework already established.

The entire structure unfolds from a single datum: the logarithmic 1-form $\eta = d\log(z_1 - z_2)$. At n points it became a differential on the Fulton–MacPherson compactification; across all genera it became the total bar differential $D_{\mathcal{A}}$; in the modular convolution algebra it became the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$. One 1-form, one relation (Arnold), one extraction (residues): a universal invariant of chiral algebras that is algebraic, computable, and complete.